18.212: Algebraic Combinatorics

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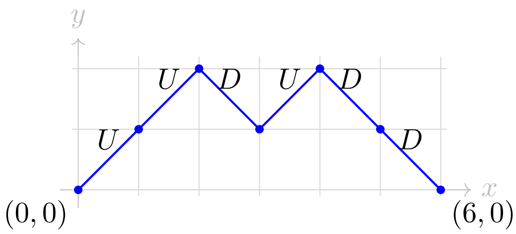
⋆ Bonus content (things left as an exercise during lecture) are be marked with a ⋆ to the left, like this

§1 Catalan numbers

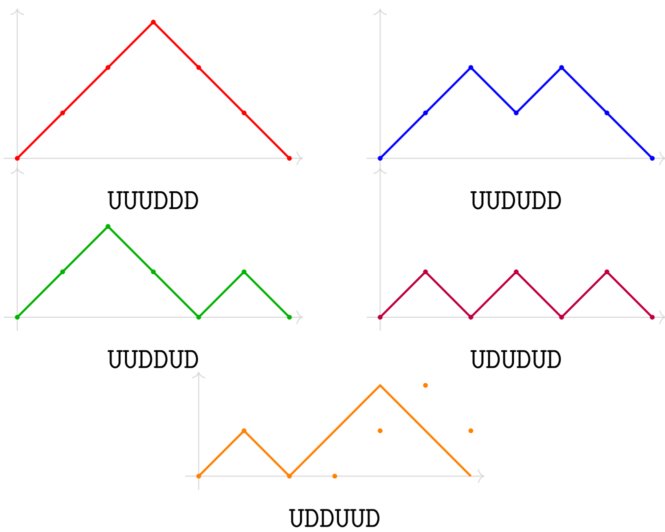
Define a **Dyck path** of length $2n$ to be a sequence $a_0 = 0, a_1, a_2, \dots, a_{2n} = 0$ such that $|a_i - a_{i+1}| = 1$ for all i. We can interpret this as a path in the xy-plane starting at $(0,0)$ and ending at $(2n, 0)$ consisting of n "up" steps and n "down" steps, such that the path never goes below the line $y = 0$.



Path never goes below $y = 0$

The **Catalan numbers** C_n count the number of **Dyck paths** of length $2n$.

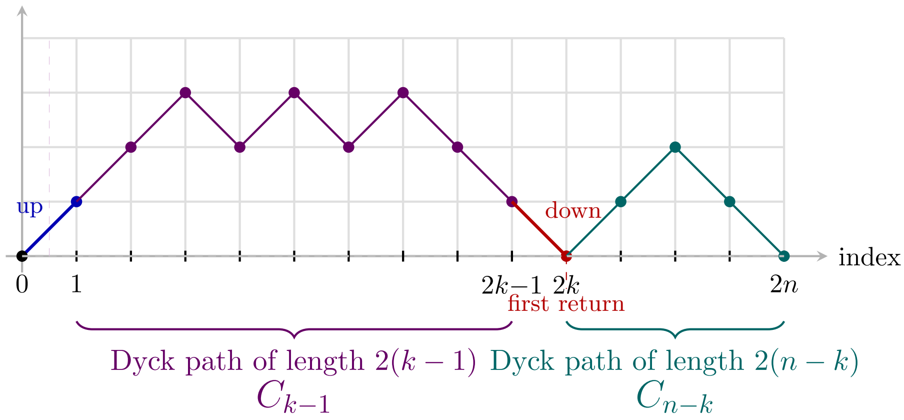
$C_3 = 5$ Dyck paths of length 6



§1.1 A formula for the Catalan numbers

We will show that the Catalan numbers are given by $C_n = \frac{1}{n+1}\binom{2n}{n}$ by writing out a recurrence and using generating functions.

We wish to count the number of Dyck paths of length $2n$. Suppose that the first time the path returns to the x-axis is at some index $2k$, for some $1 \le k \le 2n$. (It is clear for parity reasons that the path can only return to the x-axis at even indices.)



Of course, the step from index 0 to 1 must be up, and the step from index $2k-1$ to $2k$ must be down. Then, as seen above, the subsequence of indices $1, \dots, 2k-1$ forms a Dyck path of length $2(k-1)$, and the sub-sequence of indices $2k, \dots, 2n$ forms a Dyck path of length $2(n-k)$. These Dyck paths are independent of each other, so we conclude the number of Dyck paths which first return to the x-axis at index $2k$ is $C_{k-1}C_{n-k}$.

We conclude

$$C_n = \sum_{k=1}^{n} C_{k-1}C_{n-k} = \sum_{k=0}^{n-1} C_k C_{n-1-k}.$$

Now define the generating function $f(x)$ with coefficients $[x^n]f(x) = C_n$, i.e. $f(x) = \sum_{x=0}^{\infty} C_n x^n$. Observe that $[x^n]f(x)^2 = \sum_{k=0}^{n} C_k C_{n-k}$; thus $[x^n]xf(x)^2 = \sum_{k=0}^{n-1} C_k C_{n-k-1}$. Since $C_0 = 1$, we conclude from our recurrence relation that

$$f(x) = xf(x)^2 + 1.$$

Using the quadratic formula to solve for $f(x)$ in terms of x, we get $f(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$. We can determine the sign to be $\frac{1-\sqrt{1-4x}}{2x}$ by recalling that f satisfies $\lim_{x\to 0} f(x) = C_0 = 1$. Finding a formula for C_n now reduces to finding $[x^n]f(x)$. We can do this using the **Generalized Binomial Theorem**:

$$(1+y)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} y^k, \qquad \text{where } \binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}.$$

Substituting, we get

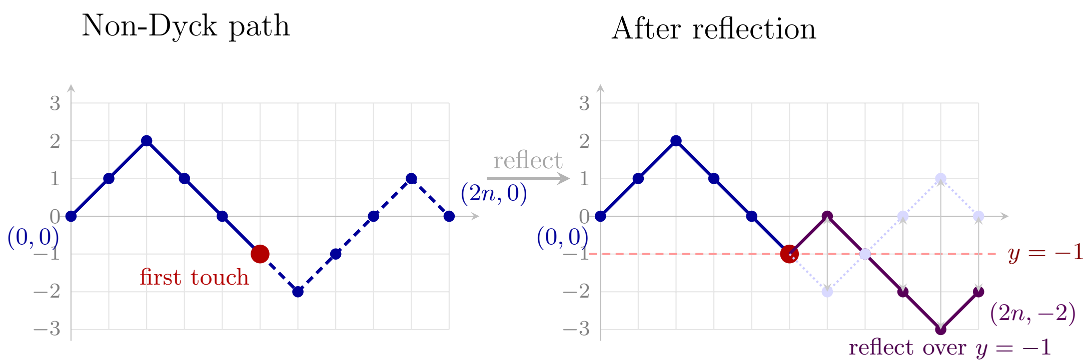
$$C_n = [x^n]f(x) = [x^{n+1}] - \frac{1}{2}(1-4x)^{1/2} = -\frac{1}{2}\binom{1/2}{n+1}(-4)^{n+1},$$

which after some arithmetic simplifies to $\frac{1}{n+1}\binom{2n}{n}$, as desired.

§1.2 Combinatorial argument via Dyck paths

We can also give a direct combinatorial proof of the formula for the Catalan numbers using the **reflection method**. We'll count the number of *non-Dyck paths*, i.e. paths from $(0,0)$ to $(2n,0)$ which *do* go below the x-axis. Since the total number of paths from $(0,0)$ to $(2n,0)$ is just $\binom{2n}{n}$, it would suffice to show that the number of non-Dyck paths is $\binom{2n}{n} - C_n = \frac{n}{n+1}\binom{2n}{n} = \binom{2n}{n-1}$. We do this by showing a bijection between non-Dyck paths and paths from $(0,0)$ to $(2n,-2)$, i.e. having $n+1$ down steps and $n-1$ up steps, the latter of which clearly there are $\binom{2n}{n-1}$.

We map non-Dyck paths to paths from $(0,0)$ to $(2n,-2)$ as follows: since a non-Dyck path goes below the x-axis, we can take the first index where it touches the $y=-1$ line, then reflect the remainder of the sequence over $y=-1$. Since the original path ends at $(2n,0)$, its corresponding path will end at $(2n,-2)$. The reverse mapping is essentially identical: one notes that any path from $(0,0)$ to $(2n,-2)$ must intersect the line $y=-1$, finds the first intersection, and then reflects the rest of the sequence back over the line.



§1.3 Combinatorial argument via cyclic shifts

We give another proof that $C_n = \frac{1}{n+1}\binom{2n}{n}$, this time using the **cycle lemma**.

Lemma (Cycle lemma)
Let $a_1, a_2, \ldots, a_m$ be a sequence of integers summing to 1. Then exactly one of the m cyclic shifts

$$a_i, a_{i+1}, \ldots, a_m, a_1, \ldots, a_{i-1} \qquad (1 \le i \le m)$$

has all of its partial sums positive.

Proof. Consider the partial sums $s_k = a_1 + a_2 + \cdots + a_k$ for $k = 1, \ldots, m$, so that $s_m = 1$. Let i be the unique index at which s_i achieves its minimum; if the minimum is achieved multiple times, take the *last* such index. (We will show this choice of i is the only one that works.)

We claim the cyclic shift starting at a_{i+1} (indices mod m) has all partial sums positive. Indeed, the kth partial sum of this shifted sequence is

$$s_{i+k} - s_i$$

where indices are taken cyclically (i.e. $s_{i+k} = s_m + s_{(i+k)-m}$ when $i + k > m$). For k such that $i + k \le m$, we have $s_{i+k} > s_i$ since i was the *last* minimiser, so $s_{i+k} - s_i > 0$. For k such that $i + k > m$, let $j = i + k - m$, so the partial sum is $s_m + s_j - s_i = 1 + s_j - s_i$. Since s_i is the global minimum, $s_j \ge s_i$, so $1 + s_j - s_i \ge 1 > 0$.

Conversely, if some other shift starting at a_{j+1} (with $j \ne i$) also had all partial sums positive, then one can verify that $s_j < s_i$, contradicting that s_i is the minimum. So exactly one shift works. $\square$

Now we use the cycle lemma to count C_n. Consider all sequences of length $2n + 1$ consisting of $(n + 1)$ copies of $+1$ and n copies of -1. There are $\binom{2n+1}{n+1} = \binom{2n+1}{n}$ such sequences, and each has total sum $+1$.

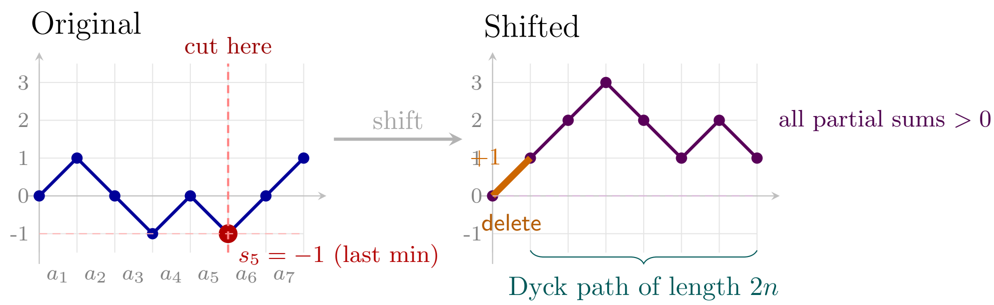
By the cycle lemma, each such sequence has *exactly one* cyclic shift whose partial sums are all positive. Conversely, every length-$(2n + 1)$ sequence of $(n + 1)$ copies of $+1$ and n copies of -1 with all partial sums positive arises this way. Moreover, distinct cyclic shifts of the same sequence are distinct sequences (since the sum is 1, no nontrivial cyclic shift can equal the original[1]). So the $\binom{2n+1}{n}$ sequences split into groups of $2n + 1$, each group contributing exactly one "good" sequence. The number of good sequences is therefore

$$\frac{1}{2n+1}\binom{2n+1}{n}.$$

Finally, observe that any good sequence (all partial sums positive) must begin with $+1$. Deleting this initial $+1$ yields a sequence $(a_2, \ldots, a_{2n+1})$ of length $2n$ with n copies of $+1$ and n copies of -1, whose partial sums satisfy $a_2 + \cdots + a_k \ge 0$ for all k—that is, a Dyck path. This is a bijection, so

$$C_n = \frac{1}{2n+1}\binom{2n+1}{n} = \frac{1}{2n+1} \cdot \frac{(2n+1)!}{(n+1)!\,n!} = \frac{(2n)!}{(n+1)!\,n!} = \frac{1}{n+1}\binom{2n}{n}.$$

[1] If $a_1, \ldots, a_m$ equals its cyclic shift by d, then the sum over any block of length d is constant, say c. Then $m/d \cdot c = 1$. Since $m = 2n + 1$ is odd and the a_i are integers, we need $d = m$ and $c = 1$, i.e. the shift is trivial.



> **Remark.** In lecture, we wondered if one could give a combinatorial argument by partitioning the paths from $(0, 0)$ to $(2n, 0)$ into $n + 1$ equally-sized classes, one of which was the set of all Dyck paths. It turns out this is possible; see this.

§1.4 Many bijections

It turns out that the Catalan numbers count many interesting objects. Below, we list several.

Claim — A **parenthesization** of n variables $x_1, \ldots, x_n$ is an assignment of a complete binary bracketing to the product $x_1 \cdot x_2 \cdots x_n$, specifying the order in which the $n-1$ pairwise multiplications are to be performed.

There are C_n valid parenthesizations of $n+1$ variables.

Example (parenthesizations)

Here are all $C_3 = 5$ parenthesizations of 4 variables:

$$((x_1x_2)x_3)x_4, \quad (x_1(x_2x_3))x_4, \quad (x_1x_2)(x_3x_4), \quad x_1((x_2x_3)x_4), \quad x_1(x_2(x_3x_4))$$

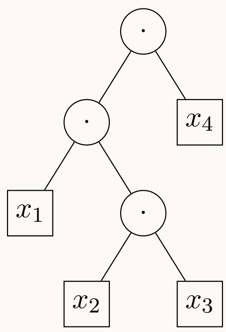
Claim — A **full binary tree** with $n+1$ leaves is a binary tree where each internal node represents one binary multiplication. We label the leaves (in left-to-right order) $x_0, \ldots, x_n$.

There are C_n full binary trees with $n+1$ leaves.

We can biject this to parenthesizations by assigning a variable to each leaf, and putting a bracket at each non-leaf node around the clauses corresponding to its left and right children.

Example (bijection between parenthesizations and trees)

Here is a binary tree that corresponds to $(x_1(x_2x_3))x_4$:



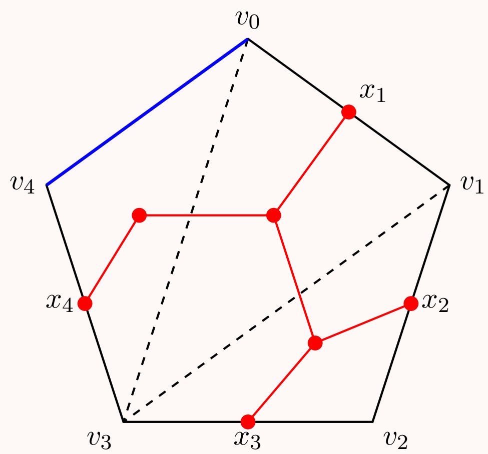
Claim — A **triangulation** of a regular $(n+2)$-gon with vertices labelled $v_0, v_1, \ldots, v_{n+1}$ is a decomposition of the polygon into n triangles by $n-1$ pairwise non-crossing diagonals.

There are C_n triangulations of an $(n+2)$-gon.

Note that a triangulation of an $(n+2)$-gon contains n triangles. We can biject triangulations to full binary trees by fixing the edge v_0v_{n+1} as the *root edge* and identifying the remaining $n+1$ polygon edges $v_0v_1, v_1v_2, \ldots, v_nv_{n+1}$ (in order) with the $n+1$ leaves of a full binary tree. Each of the n triangles corresponds to an internal node: the triangle incident to the root edge is the root, and each triangle's two non-parent edges determine its children (a polygon edge yields a leaf; a diagonal leads to a subtree on the other side).

Example (bijection between trees and triangulations)

The tree in the previous example corresponds to this triangulation of a pentagon:



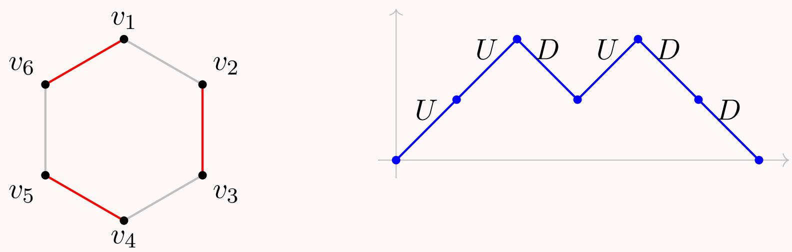
Claim — A **non-crossing matching** of a regular $2n$-gon with vertices labelled $v_1v_2\dots v_{2n}$ is a matching of the vertices, i.e. a set of n edges such that each vertex is on one edge, where none of the edges intersect.

There are C_n non-crossing matchings of a $2n$-gon.

⋆ We can biject non-crossing matchings to Dyck paths as follows: walk along the vertices $v_1, v_2, \dots, v_{2n}$ in order. At each vertex, if its partner has *not* yet been visited, record an up step U; if its partner *has* already been visited, record a down step D. The non-crossing condition ensures the result is a valid Dyck path (i.e. never dips below the x-axis).

Example

Consider the non-crossing matching $\{v_1v_6,\ v_2v_3,\ v_4v_5\}$ of a hexagon. Walking: v_1 (partner v_6 unseen, U), v_2 (partner v_3 unseen, U), v_3 (partner v_2 seen, D), v_4 (partner v_5 unseen, U), v_5 (partner v_4 seen, D), v_6 (partner v_1 seen, D). This gives the Dyck path $UUDUDD$.



Claim — A **non-crossing set partition** of a regular n-gon with vertices labelled $v_1 v_2 \dots v_n$ is a partition of the vertices into k (unordered) sets

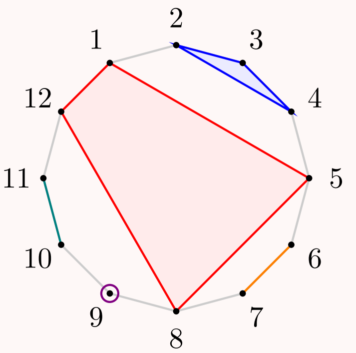
$$\{v_1, v_2, \dots, v_n\} = S_1 \sqcup S_2 \sqcup \cdots \sqcup S_k,$$

such that each vertex is in exactly one set, and the convex hulls of the sets are all nonintersecting.

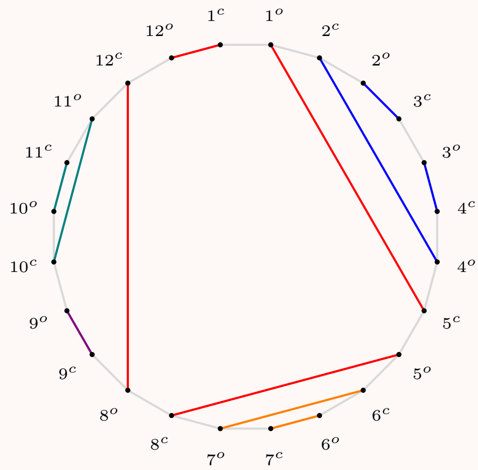
⋆ We can biject non-crossing set partitions to non-crossing Dyck paths by "expanding" each of the n vertices to a pair of "close" and "open" vertices, then within each set $S_i = \{s_{i,1}, s_{i,2}, \dots, s_{i,k}\}$ of vertices, then connecting the open-vertex of each $s_{i,j}$ to the close-vertex of $s_{i,j+1}$, looping back around to connect the open-vertex of $s_{i,k}$ with close-vertex of $s_{i,1}$. (Note that for a singleton set, this just matches the open and close vertex.)

Example (bijection from partition of 12-gon to matching of 24-gon)

Consider the non-crossing partition $\{1, 5, 8, 12\} \sqcup \{2, 3, 4\} \sqcup \{6, 7\} \sqcup \{9\} \sqcup \{10, 11\}$ of a 12-gon. Expanding each vertex v_i to a pair $v_i^{\mathrm{c}}, v_i^{\mathrm{o}}$ gives a 24-gon.



Partition of 12-gon

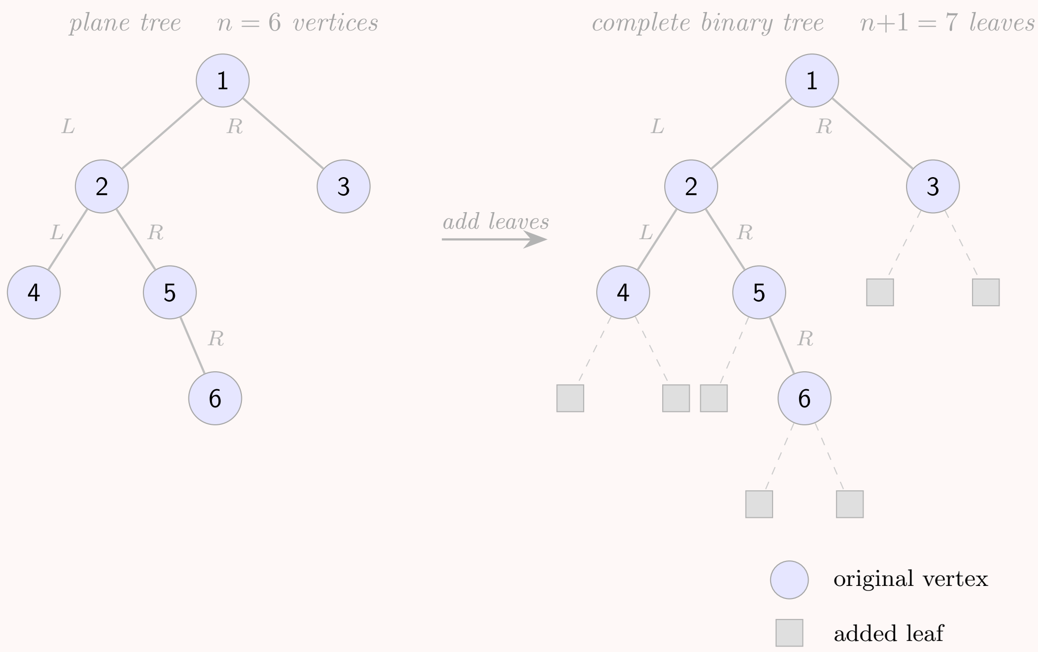


Non-crossing matching of 24-gon

Claim — A **plane tree** is a tree drawn in the plane where each vertex can have left- and right-children. (In particular, non-leaf vertices are not required to have both children.) There are C_n plane trees with n vertices.

We can biject plane binary trees on n vertices to complete binary trees with $n+1$ leaves by adding two children to each leaf node, and one leaf per node with only one child.

Example



The reverse bijection, from complete binary trees wth $n+1$ leaves to plane binary trees with n vertices, is simply to remove all the leaves.

§1.5 Stack-sortable permutations

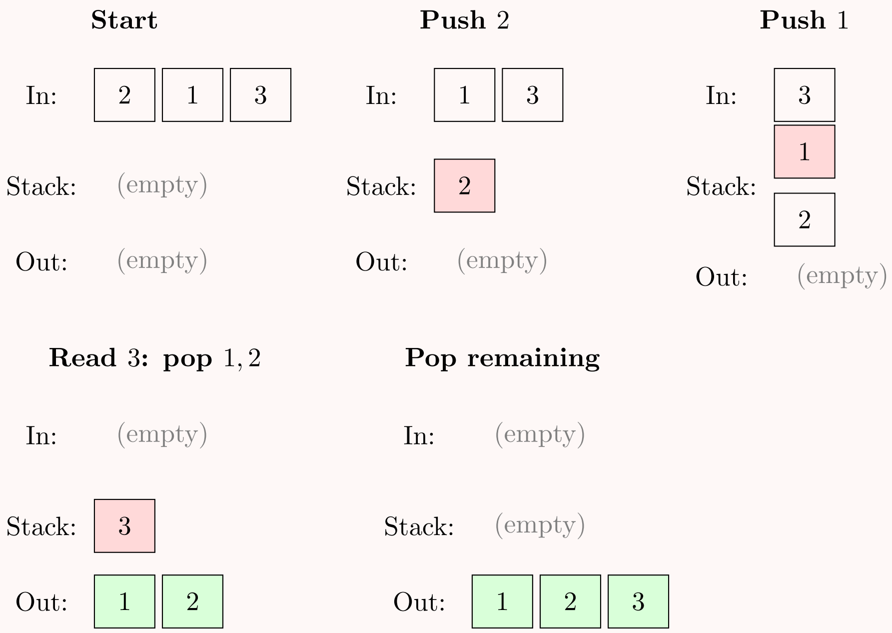
Definition. For permutations $w \in S_n$ and $\sigma \in S_m$ (for $m \le n$), say that w is **σ-avoiding** if there is no subsequence of (not necessarily consecutive!) indices $1 \le i_1 < i_2 < \cdots < i_m \le n$ such that $w(i_1), w(i_2), \ldots, w(i_m)$ are in the same relative order as $\sigma(1), \sigma(2), \ldots, \sigma(m)$. That is, w contains no subsequence of length m that is order-isomorphic to σ.

We study 231-avoiding permutations in this section. It turns out these have an equivalent characterization, in terms of how they may be constructed:

Definition. A permutation $w \in S_n$ is **stack-sortable** if it can be sorted into the identity using a single stack, via the following procedure: read $w(1), w(2), \ldots, w(n)$ left to right. At each step, either push the current element onto the stack, or pop the top of the stack to the output. The permutation is stack-sortable if some sequence of such operations produces the output $1, 2, \ldots, n$.

Example

Here's how to stack-sort the permutation 213:



Claim — A permutation $w \in S_n$ is stack-sortable if and only if it is 231-avoiding. In particular, there are C_n stack-sortable permutations in S_n.

⋆

Proof. Note that if it is possible to stack-sort a permutation, we *must* do it via the following "greedy" algorithm:

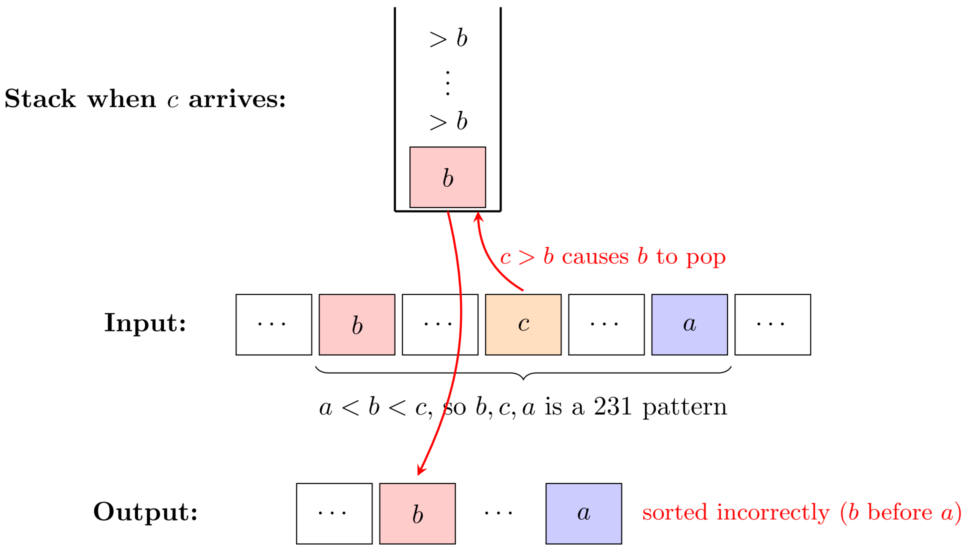
> **Algorithm** — Upon seeing the next element x in the permutation, pop all elements out of the stack that are less than x (to the output), then push x onto the stack. (When no elements remain in the input, pop everything remaining.)

This is forced because the stack must always be strictly decreasing from top to bottom: if we were to push x on top of some $y < x$ without first popping y, then y would be trapped below x and would be output after x, but $y < x$ means y must be output first in any valid sort. So we must pop all elements less than x before pushing x.

Forward direction. Suppose w contains a 231 pattern: there exist indices $i < j < k$ with $w(k) < w(i) < w(j)$. When we read $w(j)$, since $w(j) > w(i)$, we are forced to pop $w(i)$ from the stack, presuming it hasn't already been popped already; regardless, $w(i)$ is output before $w(k)$ is even read from the input. But $w(k) < w(i)$, so $w(k)$ should precede $w(i)$ in the sorted output, contradiction.

Reverse direction. We show the contrapositive: if the greedy algorithm fails, then w contains a 231 pattern. Suppose the algorithm yields an unsorted sequence, so there exist values $a < b$ with b appearing before a in the output. Consider the moment b is popped from the stack. At this moment:

1. The stack is strictly decreasing from top to bottom, so every element remaining on the stack is $> b > a$. Hence a is not on the stack.
2. Since a has not yet been output (it comes after b) and is not on the stack, a must still be unread in the input. In particular, a appears *later* in w than b.
3. Let c be the input element whose arrival caused b to be popped. Then $c > b$ (since only elements larger than the stack top will cause it to pop), and c appears after b but before a in w.



So in w, the values b, c, a appear in this order with $a < b < c$, thus forming a 231 pattern in w. □

⋆ The attentive reader will have noticed that we have constructed two sets of bijections:

- Dyck paths, $2n$-gon non-crossing matchings, n-gon non-crossing partitions,
- parenthesizations, full binary trees with $n+1$ leaves, plane binary trees with n vertices, triangulations of an $(n+2)$-gon

and thus we're still missing a link. Let's fix that.

1. Map ϕ from Dyck paths $a_1, a_2, \ldots, a_{2n}$ to plane binary trees: we will construct this recursively. Have ϕ map the empty sequence to the empty tree. Take the last index i for which the Dyck path returns to the origin; then $a_1, \ldots, a_i$ and $a_{i+2}, \ldots, a_{2n-1}$ are both Dyck paths (note that a_{i+1} is an up-step and a_{2n} is a down-step). Now, let $\phi(a_1, \ldots, a_{2n})$ be the plane binary tree where the root node has left subtree $\phi(a_1, \ldots, a_i)$ and right subtree $\phi(a_{i+2}, \ldots, a_{2n-1})$. Note that either of these subtrees is allowed to be empty, in which case there is the root node simply has no left/right child.
2. Map ψ from plane binary trees T to Dyck paths: we will again construct this recursively. Letting the left subtree of the root be T_1 and right subtree be T_2, we construct the sequence $\phi(T_1)U\phi(T_2)D$.

§1.6 Queue-sortable permutations

Definition. A permutation $w \in S_n$ is **k-queue-sortable** if it can be sorted into the identity permutation by k consecutive passes through a FIFO queue. In each pass, we read the elements of the current sequence from left to right; at each step, we may either enqueue the current element onto the back of the queue, or dequeue the front element of the queue and append it to the output sequence.

> **Claim** — A permutation w is k-queue sortable if and only if it avoids the permutation $(k+1, \ldots, 1)$.

> **Claim** — For any permutation $\sigma \in S_3$, the number of σ-avoiding permutations is C_n.

We will leave these claims unproved, though this is not the last time we will see them!

§2 Young tableaus

Definition. A **Young Tableau** of size n is a partition $\lambda = (\lambda_1, \ldots, \lambda_k)$, where $\lambda_1 \geq \cdots \geq \lambda_k$ (i.e. the partition is unordered), and $n = \lambda_1 + \cdots + \lambda_k$.

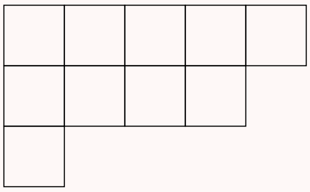
Definition. A **Standard Young Tableau** of size n is an assignment of the numbers $1, \ldots, n$ to the boxes in a Young Tableau, such that the entries in each row and each column are increasing.

We can draw out Young Tableaus as a partial grid of boxes as shown below, such that the number of boxes in each row forms the elements of the partition:

Example

The YT for partition $(5, 4, 1)$, and an SYT on that YT.

YT for $(5, 4, 1)$



An SYT on $(5, 4, 1)$

1	2	5	7	9
3	4	8	10	
6				

It turns out that SYT's generalize some notion of the Catalan numbers:

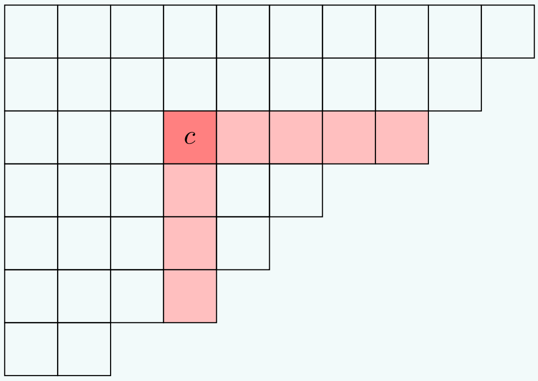
Claim — There are C_n SYT's on a $n \times 2$ YT.

We can show this by bijecting the SYT's to a Dyck path: the first row consists of all indices of up-steps, and the second row consists of all indices of down-steps. The condition that the numbers increase across each row means that each row represents an unordered set of indices, and the condition that the the numbers increase down each column means that the Dyck path we construct never goes below the x-axis.

It turns out that we can write a formula for the number of SYT's for any YT.

Theorem (Hook length formula)

For a YT λ, let f_λ be the number of SYT's with shape λ. For each box b in λ, define $\text{hook}_\lambda(b)$ to be the set of boxes either directly below or directly to the right of b, including b itself.



We will use $h(b)$ to denote the number of boxes in the hook of b, which we call the **hook length** of b. Then,

$$f_\lambda = \frac{n!}{H(\lambda)},$$

where $H(\lambda) = \prod_{b \in \lambda} h(b)$.

Here's a fake proof of the hook length theorem: there are $n!$ total ways to arrange numbers inside a YT, regardless of whether they are properly ordered. Now, for each box b, the probability its number is less than all the other boxes in its hook is $\frac{1}{h(b)}$. Note that an SYT is just an arrangement where all boxes satisfy this property, so the fraction of arrangements which is an SYT is just $\frac{1}{H(\lambda)}$ by multiplying all these probabilities together. (This proof is obviously fake, because it assumes that each event "box b is the minimum in its hook".)

Let us now present an actually-correct proof.

Proof. We will show that $f_\lambda = \frac{n!}{H(\lambda)}$ for all values of λ by showing that they both satisfy the same recurrence relation.

In particular, let a "corner" c of YT λ be a box of λ without other boxes below or to the right of it, and let $\lambda \setminus c$ be the YT where c has been removed from λ. We then have $f_\lambda = \sum_c f_{\lambda \setminus c}$ (where the sum is taken over all corners): the highest number n must be at some corner, and there are $f_{\lambda \setminus c}$ SYT's where n is at corner c.

Since $f_\lambda = \frac{n!}{H(\lambda)} = 1$ for a 1×1 YT, once we prove that $\frac{n!}{H(\lambda)}$ satisfies the same recurrence relation, we can conclude. Hence, we will devote the remainder of the proof to showing

$$\frac{n!}{H(\lambda)} = \sum_{\text{corner } c} \frac{(n-1)!}{H(\lambda \setminus c)},$$

which rearranges to

$$\sum_{\text{corner } c} \frac{H(\lambda)}{nH(\lambda \setminus c)} = 1.$$

We will show this sum is 1 by showing that $\frac{H(\lambda)}{nH(\lambda \setminus c)}$ is actually a probability measure over the set of corners:

Claim — Define a **hook walk** to be the following process on λ:

1. Begin a sequencfjlsidjfe at c_1 the top-left cell;
2. At each step from cell c_k, proceed to cell c_{k+1} by choosing a random cell in the hook of c_k (excluding c_k itself);
3. Terminate upon reaching a corner.

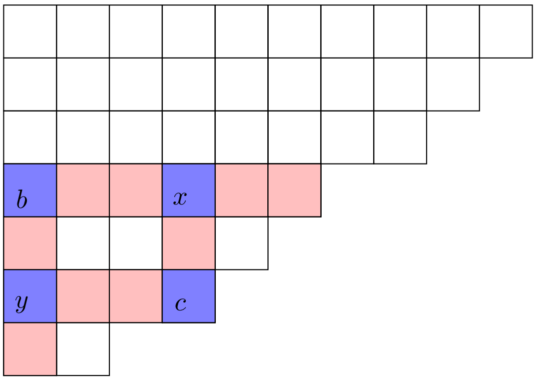
The probability of the hook walk terminating at corner c is exactly $\frac{H(\lambda)}{nH(\lambda \backslash c)}$.

Proof. Note that the probability the hook walk moves from c_k to any c_{k+1} is simply $\frac{1}{h(c_k)-1}$. Thus, the desired probability is

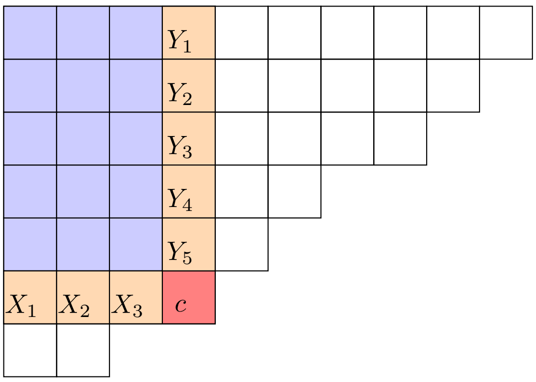
$$\sum_{\text{hook walk } b_1,\ldots,b_k,c} \left(\prod_{i=1}^{k} \frac{1}{h(b_i)-1} \right).$$

We can interpret this as a weighted sum over all the hook walks, where we assign the weight of $\frac{1}{h(b)-1}$ to all boxes b, and then assign the weight of each walk as the product of the weights of all its boxes.

Note that every hook walk that ends at c must stay within the **cohook** of c, the rectangle between the top-left corner of the Dyck path and the corner c itself. Also note that if boxes b, x, y, c form a rectangle in the YT as shown below, we have $h(b) + h(c) = h(x) + h(y)$.



Thus, for convenience, we will label the boxes lining the bottom edge of the antihook of c as $X_1, \ldots, X_k$, the boxes lining the right edge of the antihook of c as $Y_1, \ldots, Y_l$, and then let $x_i = h(X_j)$, $y_i = h(Y_j)$.



We can now refer to boxes in the hook walk by their xy-coordinates (i, j); then $h(i, j)-1 = x_i + y_i$, and our desired sum is now

$$\sum_{\text{hook walk } (i_1,j_1),(i_2,j_2),\ldots,(i_m,j_m)} \left(\prod_{p=1}^{m} \frac{1}{x_{i_p} + y_{j_p}} \right).$$

This sum is still difficult to evaluate; let us first find a formula for the weighted sum over lattice walks (where each step is to the box directly to the right or down), then use it to find the weighted sum for hook walks.

Lemma

For any lattice rectangle with bottom edge having hook lengths $x_1, \ldots, x_k$ and right edge having hook lengths $y_1, \ldots, y_l$, we have

$$\sum_{\text{lattice walk } (i_1,j_1),(i_2,j_2),\ldots,(i_m,j_m)} \left(\prod_{p=1}^{m} \frac{1}{x_{i_p} + y_{j_p}} \right) = \prod_{i=1}^{k} \frac{1}{x_i} \cdot \prod_{j=1}^{l} \frac{1}{y_j}.$$

Proof. We proceed by induction on the dimensions of the rectangle. For the base case of a $1 \times n$ rectangle, every lattice walk consists of moving right n times, giving the sum $\prod_{i=1}^{n} \frac{1}{x_i}$, which matches the formula.

For the inductive step, consider an $k \times l$ rectangle. Any lattice walk either comes from an $(k-1) \times l$ rectangle (by taking a step down) or from an $k \times (l-1)$ rectangle (by taking a step right). Applying the inductive hypotheses for both cases, and multiplying in the extra $\frac{1}{x_k + y_l}$ term corresponding to the bottom-right corner for all sequences, we get a sum of

$$\begin{aligned} \frac{1}{x_k + y_l} \left(\prod_{i=1}^{k-1} \frac{1}{x_i} \cdot \prod_{j=1}^{l} \frac{1}{y_j} + \prod_{i=1}^{k} \frac{1}{x_i} \cdot \prod_{j=1}^{l-1} \frac{1}{y_j} \right) &= \prod_{i=1}^{k} \frac{1}{x_i} \cdot \prod_{j=1}^{l} \frac{1}{y_j} \left(\frac{x_k}{x_k + y_l} + \frac{y_l}{x_k + y_l} \right) \\ &= \prod_{i=1}^{k} \frac{1}{x_i} \cdot \prod_{j=1}^{l} \frac{1}{y_j}. \end{aligned}$$

□

Now, our original sum can be expressed by summing over all possible rectangular subgrids of the cohook ending at corner c:

$$\begin{aligned}
&\sum_{\text{hook walk ending at } c} \left(\prod_{\text{box } b \text{ in walk}} \frac{1}{h(b)-1} \right) \\
&= \sum_{\substack{\text{subgrid } S \subseteq \text{cohook}(c) \\ \text{containing top-left and } c}} \left(\sum_{\text{lattice walk over } S} \left(\prod_{\text{box } (i,j) \text{ in walk}} \frac{1}{x_i + y_j} \right) \right) \\
&= \sum_{\substack{\text{subgrid } S \subseteq \text{cohook}(c) \\ \text{containing top-left and } c}} \left(\prod_{i \in S} \frac{1}{x_i} \prod_{j \in S} \frac{1}{y_j} \right) \\
&= \left(\sum_{I \subseteq \{1,\dots,k\}} \prod_{i \in I} \frac{1}{x_i} \right) \left(\sum_{J \subseteq \{1,\dots,l\}} \prod_{j \in J} \frac{1}{y_j} \right) \\
&= \prod_{i=1}^{k} \left(1 + \frac{1}{x_i} \right) \prod_{j=1}^{l} \left(1 + \frac{1}{y_j} \right),
\end{aligned}$$

where the last equality follows just by factoring. Since $x_i = h(X_i)$ and $y_j = h(Y_j)$, and since $h(X_i) = (\text{boxes to right of } X_i) + 1$ and $h(Y_j) = (\text{boxes below } Y_j) + 1$, we have $\prod_i \left(1 + \frac{1}{x_i}\right) \prod_j \left(1 + \frac{1}{y_j}\right) = \frac{H(\lambda)}{n \cdot H(\lambda \backslash c)}$ by a direct computation involving hook lengths: deleting a corner c from a YT λ decreases the hook lengths of the cells $X_1, \dot{;} X_k$ and $Y_1, \dots, Y_l$ by 1, and does not change any other hook lengths.

This completes the proof that the probability of the hook walk terminating at corner c is exactly $\frac{H(\lambda)}{n \cdot H(\lambda \backslash c)}$. □

Since the probabilities sum to 1 over all corners c, we have shown that $\sum_{\text{corner } c} \frac{H(\lambda)}{n \cdot H(\lambda \backslash c)} = 1$, which establishes our desired recurrence relation. Combined with the base case $f_\lambda = 1$ for a 1×1 tableau, this completes the proof of the hook length formula □

§2.1 The RSK correspondence

We will show a bijection between:

- permutations $w \in S_n$;
- identically-shaped pairs (p, q) of SYT's having n boxes.

For a YT λ, let $\lambda_1(\lambda)$ be the length of the longest increasing subsequence, and of the first row of a YT λ. For a permutation w, let $\lambda_1(w)$ be the length of the longest increasing subsequence in w. It turns out that our correspondence will have the property that it preserves λ_1 between permutations and YT shapes.

§2.2 A special case

Recall from earlier (though we didn't actually prove this) that the number of 123-avoiding permutations is equal to the nth Catalan number. Recall that the number of SYT's with shape $n \times 2$ is **also** the nth Catalan number: a bijection with Dyck paths is that all the numbers in the first column are the indices of the up-steps, and the numbers in the second column are indices of down-steps.

Hence, if the correspondence we claimed to exist above is real, the number of pairs (p, q) of SYT's of size n having shape λ where $\lambda_1(\lambda) \leq 2$ would be equal to the number of permutations $w \in S_n$ with $\lambda_1(w) \leq 2$, which is the number of 123-avoiding permutations, which is the nth Catalan number, which is the number of $n \times 2$ SYT's.

We can actually easily construct an explicit bijection between the first and the last objects! Given two SYT's (p, q) with the same L-shape, we construct a $n \times 2$ SYT from it by:

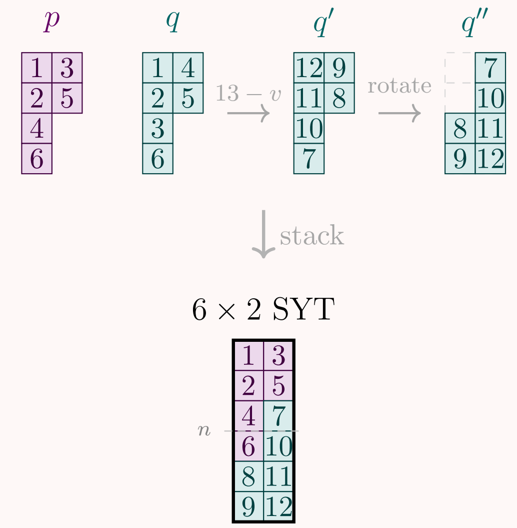
1. replace the value v in each box of q with $2n + 1 - v$;
2. rotate q around by 180°;
3. stack p on top of q.

Note that this mapping can be reversed: given a $n \times 2$ SYT, we can construct (p, q) by:

1. take p to be the subset of the boxes which are at most n;
2. take q as the rest of the boxes which are at least n, rotated 180°;
3. for each value v of a box in q, replace it with $(2n + 1) - v$.

Another interesting corollary of the correspondence is that $\sum_{\text{SYT } \lambda} f(\lambda)^2 = n!$.

Example



§2.3 Schensted Insertion Algorithm

We give an algorithm that, from a permutation $w \in S_n$, constructs a pair of identically-shaped SYT's (P, Q).

Algorithm — Building the P-Tableau: Process the elements $w_1, w_2, \ldots, w_n$ one at a time. To **insert** element x into the current tableau:

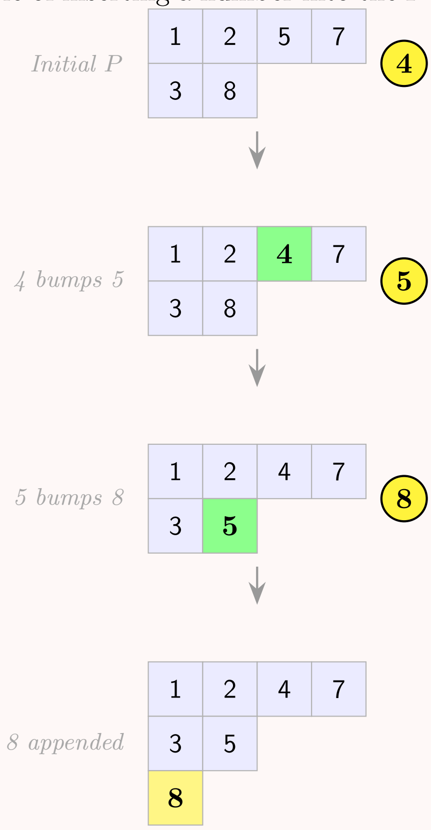
1. Find the first (leftmost) element y in row 1 which is greater than x.
2. If such y exists, replace y with x (we say x *bumps* y), and then recursively insert y into row 2 (i.e. perform this insertion algorithm again on y while ignoring row 1)
3. If no such y exists, **append** x to the end of the first row and terminate.

This bumping chain propagates downward through rows until some element is appended at the end of a row with no strictly-larger successor, creating exactly one new cell.

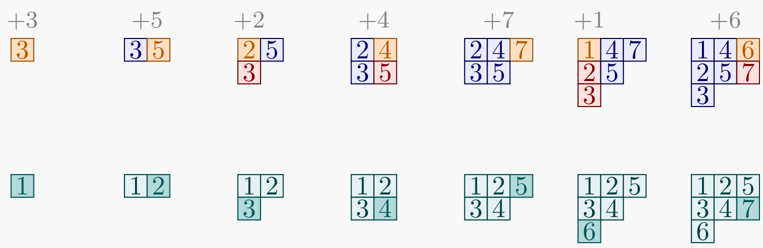
Building the Q-Tableau: Whenever the i-th element w_i is inserted and creates a new cell at position (r, c) in P, place the integer i into position (r, c) in Q.

Example

Here is an example of inserting a number into the P-tableau, which recursively bumps other numbers:



Remark. The example we did in class was this, with $w = 3524716$:



Note that this is actually a rather misleading example, because it seems to suggest you insert entries by finding the correct column, then pushing the entirety of that column down by one. In general, as you can see in the example above, that is *not* correct.

§2.4 Properties of the Schensted shape

Call the **Schensted shape** of a permutation w the shape of the tableaus (p, q) that we get from the RSK correspondence. Here are some properties of the Schensted shape:

Claim — Let λ_1 be the length of the first row of λ. Then λ_1 is also the length of the longest increasing subsequence of w.

Proof. For each $1 \le i \le \lambda_1$, let B_i be the set of numbers that are placed into the ith box in row 1. Note that an element may be placed in the ith box of row 1, and later bumped into row 2; it still belongs to B_i. Every column of row 1 is occupied at the end of the algorithm, so each B_i is non-empty.

> **Remark.** Note that B_i is *not* just the ith column of P, because numbers can change columns when we run the SIA.

Lemma

The numbers in B_i occur in decreasing order in w.

Proof. Let $x, x' \in B_i$ with x appearing before x' in w; we show $x > x'$. At the moment x' is inserted into the tableau, let z be the current occupant of column i in row 1. Since x' is placed in column i, it bumps z, so $x' < z$ by the definition of bumping. It remains to show $z \le x$. Observe that each time a value is placed in column i of row 1, it strictly displaces (and is therefore strictly less than) its predecessor there. Hence the sequence of values occupying column i over the course of the algorithm is strictly decreasing. Since x occupied column i at some point before x' was inserted, the current occupant z satisfies $z \le x$. Therefore $x' < z \le x$, as required. □

This implies the length of any increasing subsequence of w is at most λ_1, because any subsequence with more than λ_1 elements would include two elements from the same set B_i by the Pigeonhole Principle, and therefore

Lemma

For each x in B_i (for $i > 1$), there exists a $y \in B_{i-1}$ such that $y < x$ and y appears before x in the permutation.

Proof. This is the y that is in the first row, $(i-1)$th column at the moment of the SIA when x is placed in column i. This y is currently in row 1, so it must have been put there originally, and thus it must be assigned to B_{i-1}. It is clear that y is before x in the permutation, because it's in the tableau at a time when x is not. □

This implies there exists a increasing subsequence of w having length λ_1, because we can take an x_{λ_1} in B_{λ_1}, then find an $x_{\lambda_1 - 1} < x_{\lambda_1}$ preceding it in w, and so on.

We conclude that the length of the longest increasing subsequence in w is exactly λ_1. □

This actually generalizes in a manner that allows us to interpret the sizes of all the rows in the Schensted shape:

Theorem (Greene)

If $\lambda = (\lambda_1, \lambda_2, \dots)$ is the Schensted shape of permutation w, then for any k, $\lambda_1 + \cdots + \lambda_k$ is the maximum size of a union of k increasing subsequences in w.

Also, if $\lambda'_1, \lambda'_2, \dots$ are the sizes of the columns of λ, then $\lambda'_1 + \cdots + \lambda'_k$ is the maximum size of a union of k decreasing subsequences in w.

Note, however, that one cannot always construct a maximal union of k increasing subsequences via the greedy algorithm (where you select the longest increasing subsequence, remove it from the permutation, then repeat). For example, consider $w = 24135$, where $\lambda_1 = 3$ and $\lambda_2 = 5$: the greedy algorithm fails if it selects the increasing sequence $2, 3, 5$ in the first round.

Also, this is true:

> **Claim —** If RSK sends permutation σ to (P, Q), then it sends the inverse σ^{-1} to (Q, P).

§3 Permutations

§3.1 Eulerian numbers

Consider the generating function for $1 + 2^a x + 3^a x^2 + 4^a x^3 + \dots$. We can evaluate such generating functions as follows:

$$\begin{aligned}
1 + x + x^2 + \cdots &= \frac{1}{1-x} \\
1 + 2x + 3x^2 + \cdots &= \frac{1}{(1-x)^2} \\
1 + 2^2 x + 3^2 x^2 + \cdots &= \left(x \cdot \frac{1}{(1-x)^2}\right)' = \frac{1+x}{(1-x)^3} \\
1 + 2^3 x + 3^3 x^3 + \cdots &= \left(x \cdot \frac{1+x}{(1+x)^3}\right)' = \frac{1+4x+x^2}{(1-x)^4},
\end{aligned}$$

and so on—in general, the nth generating function is $f_n(x) = \sum_{j=0}^{\infty}(j+1)^n x^{j-1} = \frac{A_n(x)}{(1-x)^{n+1}}$, where $A_n(x)$ is a polynomial in x with integer coefficients, satisfying the recurrence relation

$$f_{n+1}(x) = \frac{d}{dx}(x f_n(x)).$$

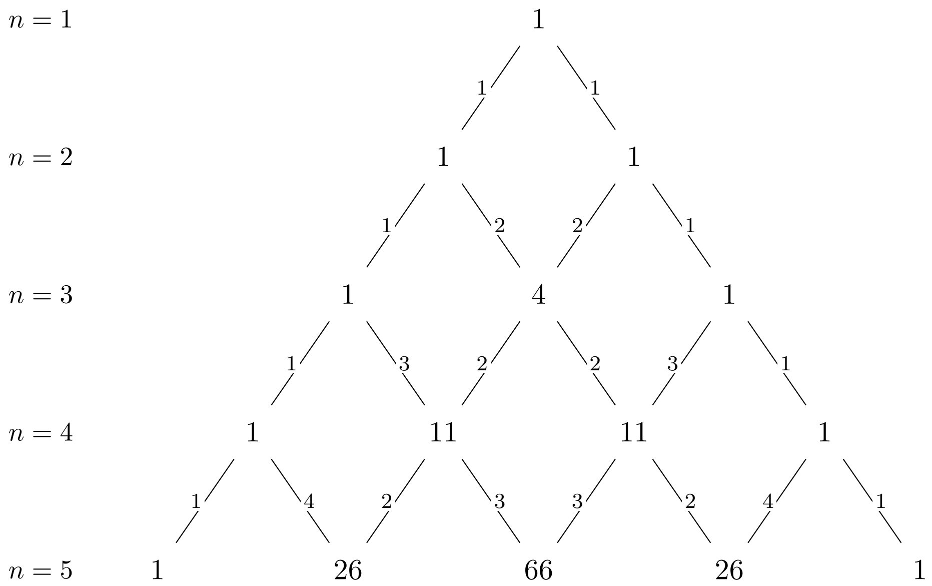
Note that this can be rewritten by clearing out the denominators as

$$A_{n+1}(x) = (1+nx)A_n(x) + x(1-x)A_n'(x).$$

Letting $A(n,k)$ denote the kth term of the polynomial A_n, we can now find the following recurrence relation:

$$\begin{aligned}
A(n+1,k) = [x^k]A_{n+1}(x) &= [x^k](1+nx)A_n(x) + [x^k]x(1-x)A_n'(x) \\
&= [x^k]A_n(x) + n[x^{k-1}]A_n(x) + k[x^k]A_n(x) - (k-1)[x^{k-1}]A_n(x) \\
&= (k+1)[x^k]A_n(x) + (n-k-1)[x^{k-1}]A_n(x) \\
&= (k+1)A(n,k) + (n-k+1)A(n,k-1).
\end{aligned}$$

We call $A(n,k)$ the **Eulerian numbers**; they can be visualized by placing the numbers in an "Eulerian triangle", where each number is the weighted sum of the two numbers above it:



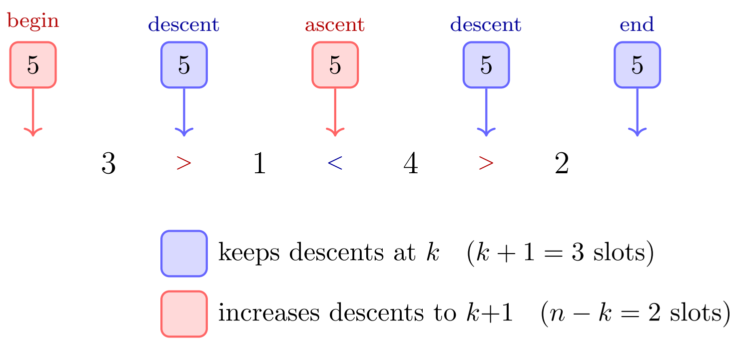
Notice that the sum of the numbers in row n is just $n!$. We'll show that this is not a coincidence:

> **Claim** — $A(n,k)$ is the number of permutations in S_n with k **descents** (adjacent indices $i, i+1$ of w for which $w(i) > w(i+1)$).

Proof. Let $A'(n,k)$ be this number of permutations. It is quite clear that $A'(n,0) = A'(n,n-1) = 1$ for all n, so all we need to do is show that $A'(n,k)$ satisfies the same recurrence relation as $A(n,k)$.

To do this, imagine we are counting some nondegenerate $A'(n+1,k)$:

- if $n+1$ is placed between some x, y such that $x > y$, then the number of descents remains the same if we remove $n+1$ to get a permutation in S_n. There are $A(n,k)$ such permutations, and $k+1$ ways to insert $n+1$ into a descent (or at the beginning).
- if $n+1$ is placed between some x, y such that $x < y$, then the number of descents increases by 1 if we remove $n+1$ to get a permutation in S_n. There are $A(n,k-1)$ such permutations, and $n-k+1$ ways to insert $n+1$ into an ascent (or at the beginning).



$\square$

Another interesting characterization of the Eulerian numbers is via the following probability distriution:

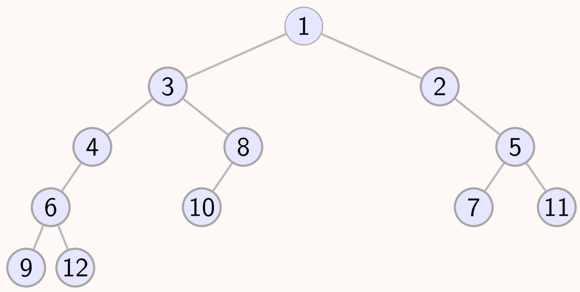
Claim — Suppose $x_1, \dots, x_n \sim U(0,1)$. Then $x = \lfloor x_1 + \cdots + x_n \rfloor$ has the probability mass function

$$\mathbb{P}[y = k] = \frac{A(n,k)}{n!}.$$

Yet another characterization of Eulerian numbers arises from binary trees. Consider "increasing" binary trees—labelings of the vertices of a plane binary tree such that the label of any child is greater than that of the parent. We first note that we can biject $w \in S_n$ to increasing binary trees on n vertices: given w, construct the tree recursively by finding the smallest element in the permutation, setting it as a parent, and construct its left and right subtrees from all the elements before and after w in the permutation.

Example

The tree we'd construct for the permutation 9 6 12 4 3 10 8 1 2 7 5 11.



Note that left edges in the tree correspond to descents and missing left edges in the tree correspond to ascents in the permutation. Thus the number of trees with k left edges is the same as the number of permutations with k descents, namely $A(n,k)$.

It turns out that for a general plane tree T, the following theorem holds:

Theorem

Let f_T be the number of increasing labelings of T, and let $h(v)$ for any vertex v of T be the number of vertices in the subtree of v. Then

$$f_T = n! \prod_v \frac{1}{h(v)}.$$

This should remind you of the hook length formula, though the proof can be done directly by induction on the size of the tree.

§3.2 Narayana Numbers

Here's a similar sequence, while we're at it: the **Narayana numbers** $N(n,k)$ are the number of Dyck paths of $2n$ steps that have k "peaks" (i.e. subsequences of adjacent up and down steps).

Example

Here are some of the Narayana numbers, drawn in a triangle:

$$\begin{array}{ccccccccccc}
 & & & & & 1 & & & & & \\
 & & & & 1 & & 1 & & & & \\
 & & & 1 & & 3 & & 1 & & & \\
 & & 1 & & 6 & & 6 & & 1 & & \\
 & 1 & & 10 & & 20 & & 10 & & 1 & \\
1 & & 15 & & 50 & & 50 & & 15 & & 1
\end{array}$$

Claim — The Naryana number $N(n,k)$ is the number of plane binary trees with n vertices and $k-1$ left edges.

⋆ This is true because the map we previously described between Dyck paths to plane binary trees actually just translates between these two properties.

From the above, we deduce that the Narayana numbers are symmetrical: $N(n,k) = N(n, n-k)$. This is not obvious from our original definition!

Also, to finish things off, the Narayana numbers have the following closed form:

Claim — The Narayana numbers are given by $N(n,k) = \frac{1}{n}\binom{n}{k}\binom{n}{k-1}$.

§3.3 Statistics on permutations

Before we proceed, let's describe three ways to write out permutations:

- you can write out the permutation, e.g. 14352.
- you can write out the permutation in "two-line" format, in a table of entries $(i, w(i))$:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 5 & 2 \end{array}$$

- you can write out the permutation as a decomposition of cycles: (1)(245)(3).

A **statistic** on a permutation is any map $\sigma : S_n \to \mathbb{R}$. We will say that two statistics σ_1, σ_2 are **equidistributed** if, for all k, there are as many permutations $w \in S_n$ for which $\sigma_1(w) = k$ as $\sigma_2(w) = k$.

Claim — The number of descents of a permutation (i.e. indices i for which $w_i > w_{i+1}$) is equidistributed to the number of records of a permutation (i.e. indices i for which $w_i > w_1, \ldots, w_{i-1}$).

Proof. We give an automorphism of S_n that maps permutations with k descents to permutations with k cycles. Write the cycle decomposition of the permutation in such a way that the largest number of each cycle comes first in the cycle, and the cycles are ordered in increasing order of their largest numbers; then erase all parentheses.

$$24513 \to (412)(53) \to 41253$$

□

Claim — The exceedance of a permutation (the number of indices i for which $w_i > i$) is equidistributed to the number of descents.

Proof. First of all, exceedance is equidistributed to antiexceedance (the number of indices i for which $w_i < i$), because the exceedance of w is the antiexceedance of w^{-1}. Now note that the automorphism we described in the previous claim turns antiexceedances into descents, and we are done. □

Claim — Define the **major index** of a permutation w to be $\sum i$ over all descents i of w, and define the **inversion count** to be the number of pairs of indices $i < j$ such that $w_i > w_j$. These two statistics are also equidistributed.

⋆ For a very short proof of this last result (about half a page), see here.

§4 Posets

A **poset** is a collection of items and a binary relation $\leq$ which satisfies

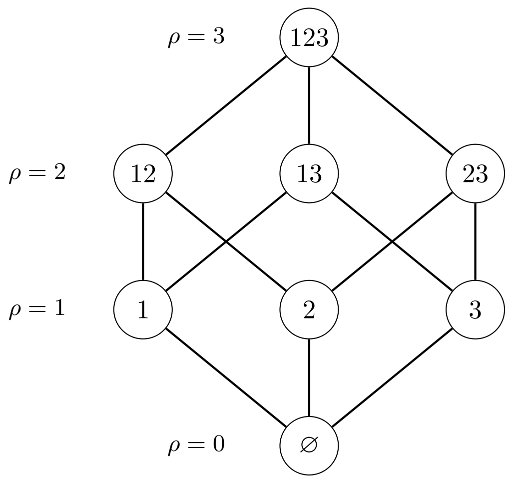
- $a \leq a$ for all a
- if $a \geq b$ and $b \geq a$, then $a = b$
- if $a \geq b$ and $b \geq c$, then $a \geq c$.

Note that two items can be incomparable, i.e. $a \leq b$ and $b \leq a$ are both not true.

Of course we can define the $<$ operator (where $a < b$ if $a \leq b$ and $a \neq b$). Now, say that b **covers** a, denoted $a \lessdot b$, if $a < b$ and there does not exist c such that $a < c < b$.

A **chain** in a poset is a sequence $a_1 < a_2 < \cdots < a_k$; call the chain **saturated** if $a_1 \lessdot < a_2 \lessdot \cdots \lessdot a_k$. An **antichain** in a poset is a set of elements, none of which can be compared.

We can now draw the *Hasse diagram* for a poset, which is the directed graph of all coverings:



We say that a poset is *ranked* if there exists a rank function ρ such that for all $a \lessdot b$, we have $\rho(b) = \rho(a) + 1$. (Thus in a Hasse diagram it would be aesthetically pleasing to draw all elements of the poset with the same rank on the same y-level.) If we let r_k be the number of elements of the poset with rank k, we say that a poset is **unimodal** if $r_0 \leq r_1 \leq \cdots \leq r_j \geq r_{j+1} \geq \cdots \geq r_n$ (where n is the highest rank in the poset), and we say that it is **rank-symmetric** if $r_i = r_{n-i}$ for all i.

A **symmetric chain decomposition** (SCD) of a ranked poset with maximum rank n (the minimum rank is conventionally 0) is a partition of the elements of the poset into disjoint saturated chains, where each one is of the form $c_0 \lessdot \cdots \lessdot c_s$ with $\rho(c_0) + \rho(c_s) = n$.

For a unimodal poset, say that it is **Sperner** if the maximum size of an antichain is the maximum rank number.

> **Claim** — If a poset has an SCD then it is rank-symmetric and Sperner.

Proof. To see that it is rank-symmetric, simply biject each element to the opposite element in its chain.

To see that it is Sperner, suppose the maximum rank is n: every chain contains an element of rank $\lfloor n/2 \rfloor$. This implies the maximum rank number is $r_{\lfloor n/2 \rfloor}$, and that no antichain can have more than $r_{\lfloor n/2 \rfloor}$ elements (else it would have two elements in a chain, which would be comparable). All the elements of rank $\lfloor n/2 \rfloor$ form an antichain, so we are done. □

> **Remark.** Of course, this is not a necessary condition; it is only sufficient.

§4.1 Sperner's Theorem

Consider the poset B_n over subsets of $[n] = \{1, \dots, n\}$, where the $\le$ operation is $\subseteq$.

- $a \lessdot b$ if $b = a \cup \{x\}$ for some $x \notin a$.
- this is a ranked poset; the rank of a is just $|a|$.
- in fact, this is a unimodal poset because $r_k = \binom{n}{k}$ is unimodal.

In this section we will prove the following:

Theorem (de Bruijn)

B_n has an SCD, and therefore is Sperner.

It is helpful to consider this as a poset over B_n, the binary hypercube $\{0, 1\}^n$, where vectors $x \in \{0, 1\}$ correspond to the subset $\{i | x_i = i\}$.

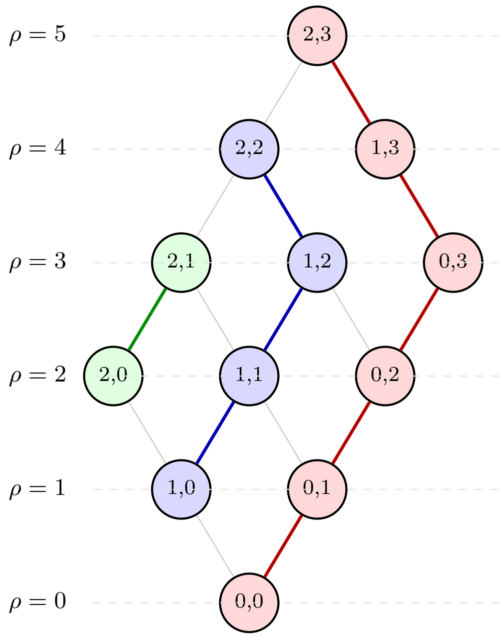
In order to prove this theorem, we will first define a product over posets: for posets P and Q, define their product $P \times Q$ to be the poset over $P \times Q$ with $\le$ relation given by

$$(p_1, q_1) \le (p_2, q_2) \text{ if } p_1 \le p_2 \text{ (in } P\text{) and } q_2 \le q_2 \text{ (in } Q\text{)}.$$

Lemma

For convenience, let $[a]$ denote a chain of length a. For any poset P and Q which have SCDs, $P \times Q$ also has an SCD.

Proof. Let's first check this is true when P is a chain $[b]$. The Hasse diagram for $[a] \times [b]$ can be given an SCD as below:



Now, suppose P has an SCD into chains $C_1, C_2, \dots, C_k$, and Q has an SCD into chains $D_1, \dots, D_\ell$. We can form an SCD of $P \times Q$ by repeating the above construction on $C_i \times D_j$ for each i, j. $\square$

We can now prove de Bruijn's theorem by noting that B_n is simply $[2] \times [2] \times \cdots \times [2]$ and repeatedly applying the above lemma.

§4.2 More facts about posets

Theorem (Dilworth)
The maximum size of an antichain is the minimum number of chains required to cover the poset.

Theorem (Minsky)
The maximum length of a chain is the mininum number of antichains required to cover the poset.

Theorem (Greene)
Consider a poset P; let $\ell_k(P)$ be the maximum size of the union of k chains of P, and $m_k(P)$ be the maximum size of the union of k antichains of P. Let $\lambda(P) = (\ell_1, \ell_2 - \ell_1, \dots)$, and $\mu(P) = (m_1, m_2 - m_1, \dots)$. Then, both $\lambda(P)$ and $\mu(P)$ are partitions, and their Young Tableaus are conjugates of each other.

Take a permutation $w \in S_n$; consider the poset P over $\{x_1, \dots, x_n\}$ where $x_i \le x_j$ if $i \le j$ and $w_i \le w_j$. Then $\lambda(P)$ in the above theorem is the Schensted shape of w, and the theorem specializes to the Greene's Theorem we saw previously.

§5 Counting partitions

§5.1 Q-numbers

Define the **q-number** $[n]_q$ to be $1+q+\cdots+q^{n-1}$. Define the **q-factorial** to be

$$[n]_q! = \prod_{k=1}^{n} [k]_q.$$

Theorem

$$[n]_q! = \sum_{w\in S_n} q^{\mathrm{inv}(w)}.$$

Proof. Induct on n. This is obviously true when $n=1$; both expressions are equal to 1. Suppose $[n-1]_q! = \sum_{w\in S_{n-1}} q^{\mathrm{inv}(w)}$. Now, take any $w\in S_n$, and consider the index $1\le k\le n$ such that $w_k = n$. Then n contributes $n-k$ inversions to w. Then remove n from w to get $w'\in S_{n-1}$.

$$\sum_{w\in S_n} q^{\mathrm{inv}(w)} = (1+q+\cdots+q^{n-1}) \sum_{w\in S_{n-1}} q^{\mathrm{inv}(w)} = [n]_q\cdot[n-1]_q! = [n]_q!.$$

$\square$

Let us define the **q-binomial coefficient** $\begin{bmatrix} n \\ k\end{bmatrix}_q = \frac{[n]_q!}{[k]_q![n-k]_q!}$. Note that this can be rewritten as

$$\begin{bmatrix} n \\ k\end{bmatrix}_q = \frac{\prod_{j=n-k}^{n-1}(1+q+q^2+\cdots+q^j)}{\prod_{j=0}^{k}(1+\cdots+q^j)} = \frac{(1-q^{n-k+1})\cdots(1-q^n)}{(1-q)\cdots(1-q^k)}.$$

These satisfy the following recurrence relation:

Claim (q-Pascal formula) — $\begin{bmatrix} n \\ k\end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1\end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k\end{bmatrix}_q.$

Proof. We compute

$$\begin{aligned}\begin{bmatrix} n-1 \\ k-1\end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k\end{bmatrix}_q &= \frac{[n-1]_q!}{[k-1]_q!\,[n-k]_q!} + q^k\,\frac{[n-1]_q!}{[k]_q!\,[n-1-k]_q!} \\ &= \frac{[n-1]_q!}{[k]_q!\,[n-k]_q!}\Big([k]_q + q^k\,[n-k]_q\Big).\end{aligned}$$

Now observe that

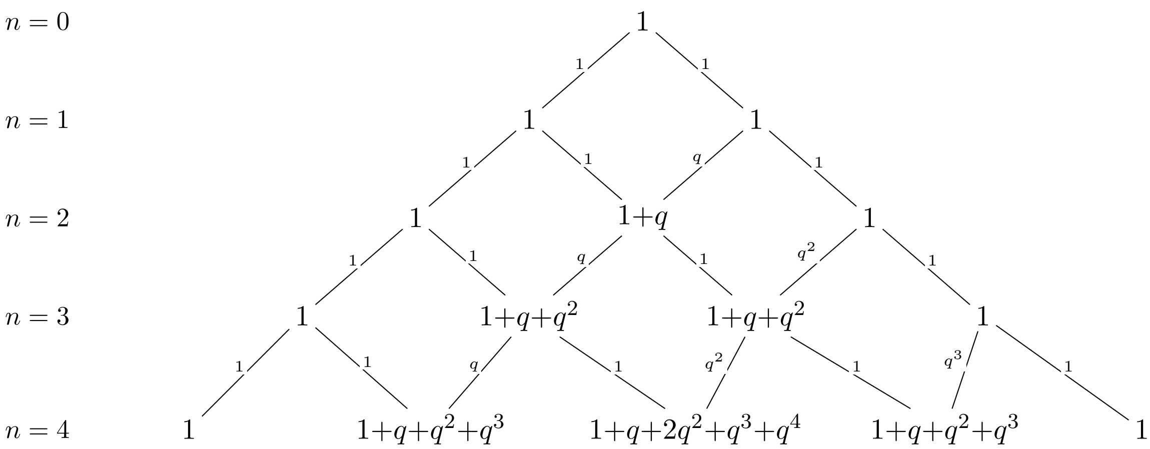
$$[k]_q + q^k\,[n-k]_q = \left(1+q+\cdots+q^{k-1}\right) + q^k\left(1+q+\cdots+q^{n-k-1}\right) = 1+q+\cdots+q^{n-1} = [n]_q.$$

Substituting, we get

$$\frac{[n-1]_q!\cdot[n]_q}{[k]_q!\,[n-k]_q!} = \frac{[n]_q!}{[k]_q!\,[n-k]_q!} = \begin{bmatrix} n \\ k\end{bmatrix}_q.$$

$\square$

This lets us draw out the q-Pascal triangle:



Note that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is always a polynomial in q.

Theorem

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|},$$

where the sum on the RHS is over all YTs λ which have height at most k and width at most $n-k$.

We can prove this directly using the q-Pascal formula for the q-binomial coefficient.

Proof by recurrence relation. Let $A(n,k,q) = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}$, so we wish to show $\begin{bmatrix} n \\ k \end{bmatrix}_q = A(n,k,q)$. The base cases $k = 1$ and $k = n-1$ are clear.

For the inductive step, we will show that

$$A(n,k,q) = A(n-1,k-1,q) + q^k A(n-1,k,q).$$

This is true by decomposing the sum λ into the YTs with height k, and the YTs with height $< k$.

- if a YT does not have height k, then it is simply a subset of $(k-1) \times (n-k)$; the sum
$$\sum_{\substack{\lambda \subseteq k \times (n-k) \\ \lambda \text{ has height } < k}} q^{|\lambda|} = \sum_{\lambda \subseteq (k-1) \times (n-k)} q^{|\lambda|} = A(n-1,k-1,q).$$
- if a YT has height k, its first column has height k. So we can just remove these k boxes and recurse:
$$\sum_{\substack{\lambda \subseteq k \times (n-k) \\ \lambda \text{ has height } k}} q^{|\lambda|} = q^k \sum_{\lambda \subseteq k \times (n-k-1)} q^{|\lambda|} = q^k A(n-1,k,q).$$

Putting this together,

$$A(n,k,q) = A(n-1,k-1,q) + q^k A(n-1,k,q),$$

which is the same recurrence relation as $\left[{n \atop k}\right]_q$ and we are done. $\square$

However, it is possible to offer a more enlightening combinatorial proof.

Definition. Suppose q is a prime power. Define the **Grassmannian** $\mathrm{Gr}(k,n,\mathbb{F}_q)$ to be the number of linear subspaces of dimension k in $(\mathbb{F}_q)^n$.

We can first find that the closed form for the Grassmannian is in fact $\left[{n \atop k}\right]_q$:

Lemma

For any integers n, k and prime power q, we have $\mathrm{Gr}(k,n,\mathbb{F}_q) = \left[{n \atop k}\right]_q$.

Proof. We construct a k-dimensional subspace of $\mathbb{F}_q^n$ by constructing a $k \times n$ matrix with full (row) rank. The first row can be chosen in $q^n - 1$ ways; anything that is not the zero vector will do. The second row can then be chosen in $q^n - q$ ways; anything that is not a scalar multiple of the first row will do. In general, the kth row can then be chosen in $q^n - q^{k-1}$ ways; anything that is not a linear combination of the first $k-1$ rows will do. Thus, the total number of such matrices is

$$(q^n - 1)(q^n - q)(q^n - q^2)\cdots(q^n - q^{k-1}).$$

However, matrices A and A' have the same row space if one can write $A = CA'$ for some invertible $k \times k$ matrix C. Thus, we have overcounted by a factor of however many such matrices there are; by setting $n = k$ above, this count is

$$(q^k - 1)(q^k - q)\cdots(q^k - q^{k-1}).$$

We conclude that the number of subspaces is

$$\frac{(q^n - 1)(q^n - q)(q^n - q^2)\cdots(q^n - q^{k-1})}{(q^k - 1)(q^k - q)\cdots(q^k - q^{k-1})} = \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

as desired. $\square$

Now we can relate the Grassmannian to our sum over YTs:

Lemma

For any integers n, k and prime power q, we have $\mathrm{Gr}(k,n,\mathbb{F}_q) = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}$.

Proof. Every k-dimensional subspace of $\mathbb{F}_q^n$ has a unique representation as a $k \times n$ matrix in reduced row echelon form (RREF). Such a matrix has pivots in some columns $1 \le j_1 < j_2 < \cdots < j_k \le n$. We count the free entries—those not forced by the RREF conditions—and establish a bijection between pivot sets and partitions $\lambda \subseteq k \times (n-k)$.

In row i, the entries in columns $1, \ldots, j_i - 1$ are zero (left of the pivot), the pivot entry is 1, and the entries in all other pivot columns $j_{i+1}, \ldots, j_k$ are zero (the "clearing" condition).

The remaining entries—those in non-pivot columns strictly to the right of j_i—are free, each ranging over all of $\mathbb{F}_q$. There are $(n-j_i)-(k-i)$ such entries in row i.

Define $c_i = j_i - i$ for $i = 1, \dots, k$. The strict increase $j_1 < \cdots < j_k$ gives $0 \le c_1 \le c_2 \le \cdots \le c_k \le n-k$, and the free entry count in row i is $(n-k)-c_i$. Now set

$$\lambda_i = (n-k) - c_{k+1-i}, \qquad i = 1, \dots, k.$$

Since (c_i) is weakly increasing, (λ_i) is weakly decreasing with $0 \le \lambda_i \le n-k$, so λ is a partition fitting inside the $k \times (n-k)$ box. This map is a bijection between valid pivot sets and such partitions (one recovers c_i and then $j_i = c_i + i$). Moreover,

$$|\lambda| = \sum_{i=1}^{k} \lambda_i = \sum_{i=1}^{k} \big((n-k) - c_{k+1-i}\big) = \sum_{i=1}^{k} \big((n-k) - c_i\big),$$

which is exactly the total number of free entries. Since each free entry independently takes any value in $\mathbb{F}_q$, the number of RREF matrices with pivot set corresponding to λ is $q^{|\lambda|}$. Summing over all partitions gives

$$\mathrm{Gr}(k, n, \mathbb{F}_q) = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}. \qquad \square$$

Note that $\begin{bmatrix} n \\ k \end{bmatrix}_q$ and $\sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}$ are both rational functions of n, k, and q. Thus, because they agree at infinitely many values of n, k, and q, they must be equal.

§5.2 The partition function

Take our formula

$$\sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|} = \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(1-q^{n-k+1}) \cdots (1-q^n)}{(1-q) \cdots (1-q^k)}.$$

Consider this as a generating function (i.e. formal power series over q). Take $n \to \infty$:

$$\sum_{\lambda \text{ with height } \le k} q^{|\lambda|} = \frac{1}{(1-q) \cdots (1-q^k)}.$$

Now take $k \to \infty$:

$$\sum_{\lambda} q^{|\lambda|} = \prod_{k=1}^{\infty} \frac{1}{1-q^k}.$$

Note that, letting $p(n)$ be the number of partitions of size n, this generating function is $\sum p(n) x^n$. This generating function has the nice combinatorial interpretation:

> **Claim —** The number of partitions of size n is the number of nonnegative integer solutions $\lambda = (\lambda_1, \lambda_2, \dots)$ to
>
> $$n = \lambda_1 + 2\lambda_2 + 3\lambda_3 + \cdots.$$

Proof. Consider λ_k to be the number of rows of length k in the partition. $\square$

Claim — The number of partitions consisting entirely of odd numbers is equal to the number of partitions having all distinct parts.

Proof. The generating function for the former is

$$\frac{1}{1-x}\cdot\frac{1}{1-x^3}\cdot\frac{1}{1-x^5}\cdots$$

and the generating function for the latter is

$$(1+x)(1+x^2)(1+x^3)\cdots.$$

Note that for any odd k,

$$(1+x^k)(1+x^{2k})(1+x^{4k})(1+x^{8k})\cdots = 1+x^k+x^{2k}+x^{3k}+\cdots = \frac{1}{1-x^k},$$

by the uniqueness of binary representations. Then, take the product over all k. □

§5.3 Euler's pentagonal number formula

Theorem

By convention set $p(0)=1$ and $p(k)=0$ for any $k<0$. Then for any n,

$$\sum_{k\in\mathbb{Z}}(-1)^k p\left(n-\frac{k(3k-1)}{2}\right)=0.$$

Note that this gives you a recurrence relation for the partition function:

$$p(n)=\sum_{k\in\mathbb{Z}\setminus\{0\}}(-1)^{k+1}p(n-\tfrac{k(3k-1)}{2}).$$

The numbers $\frac{k(3k-1)}{2}$ are known as the **pentagonal numbers**.

Proof. Recall the generating function for partition numbers:

$$\sum_n p(n)x^n=\prod_{k=1}^{\infty}\frac{1}{1-q^k},$$

Viewing the sum in the theorem as a convolution on p, we wish to show

$$\left(\prod_{k=1}^{\infty}\frac{1}{1-q^k}\right)\left(\sum_{k\in\mathbb{Z}}(-1)^k x^{k(3k-1)/2}\right)=1,$$

or equivalently

$$\prod_{k=1}^{\infty}(1-x^k)=\sum_{k\in\mathbb{Z}}(-1)^k x^{k(3k-1)/2}.$$

One can see that the LHS is

$$\sum_{\lambda\text{ with distinct parts}}(-1)^{\#\text{ parts}}x^{|\lambda|},$$

so it would suffice to show that, for all n, the difference between the number of partitions λ with $|\lambda| = n$ and all distinct parts where λ has an even number of parts, and an odd number of parts, is

$$\begin{cases} (-1)^k & \text{if } n = \frac{k(3k-1)}{2} \text{ for some } k \in \mathbb{Z}; \\ 0 & \text{otherwise} \end{cases}.$$

We prove this by constructing an (partial) map on partitions λ with distinct parts: Let the parts of λ be $\lambda_1 > \lambda_2 > \cdots > \lambda_k$.. Let ℓ be the largest index such that $\lambda_\ell = \lambda_1 - \ell + 1$.

- If $\ell < \lambda_k$, construct the partition λ' as

$$\lambda_1 - 1 > \cdots > \lambda_\ell - 1 > \lambda_{\ell+1} > \cdots > \lambda_k > \ell.$$

- If $\ell \geq \lambda_k$, construct the partition λ' as

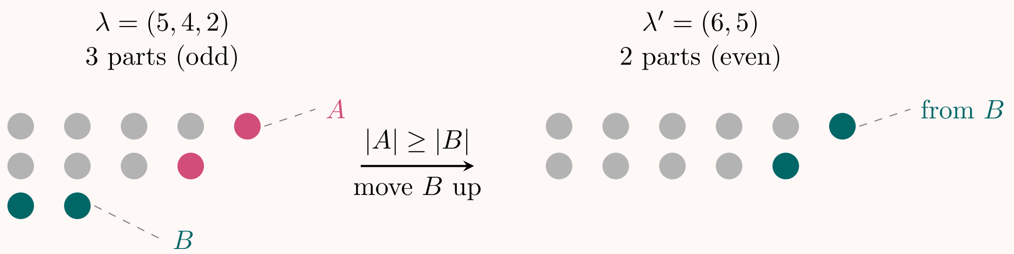
$$\lambda_1 + 1 > \cdots > \lambda_{\lambda_k+1} + 1 > \lambda_{\ell+1} > \cdots > \lambda_{k-1}.$$

This has a pleasant geometric interpretation when λ is visualized via Ferrers diagram: Let A be the set of dots on the topmost diagonal, and let B be the set of dots on the bottommost row.

- If $|A| < |B|$, construct λ' by moving A to the last row, below B.
- If $|A| \geq |B|$, construct λ' by moving B on top of the diagonal, below A.

Example

Consider $\lambda = (5, 4, 2)$, shown below with the diagonal A marked ● and the bottom row B marked ●. Since $\lambda_1 = 5$ and $\lambda_2 = 4 = 5 - 1$ but $\lambda_3 = 2 \neq 3$, we have $\ell = |A| = 2$. The bottom row has $|B| = \lambda_3 = 2$, so $|A| \geq |B|$ and we move B onto the diagonal to obtain $\lambda' = (6, 5)$.

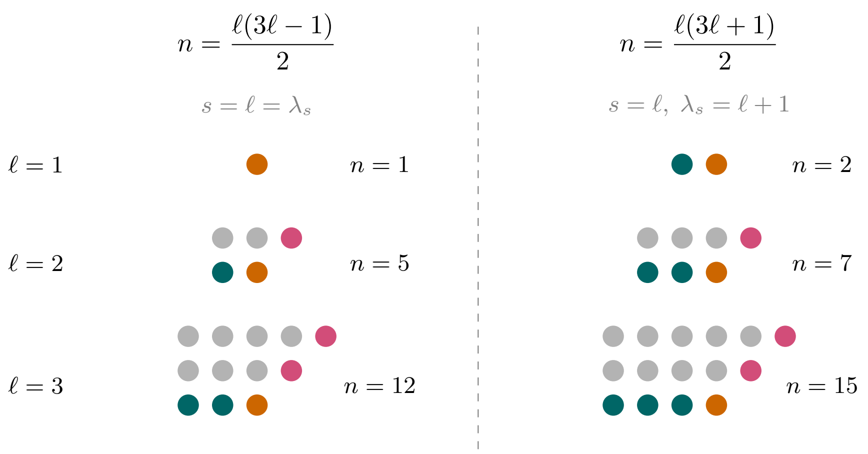


We now check when this involution is *not* well-defined. The map can fail only when A and B share their corner dot, which happens precisely when $|A| = \ell$ and $|B| = \lambda_s$ satisfy one of:

1. $\ell = \lambda_s$ and the number of parts $s = \ell$;
2. $\ell + 1 = \lambda_s$ and the number of parts $s = \ell$.

In case (1), trying to move B up would destroy the shared corner dot of A; and trying to move A down would produce a new row of size $\ell = \lambda_s$, duplicating an existing part. In case (2), $|A| < |B|$, and moving A down produces a new row of size $\ell = \lambda_s - 1$, but the resulting partition $(\lambda_1 - 1, \ldots, \lambda_\ell - 1, \lambda_{\ell+1}, \ldots, \lambda_s, \ell)$ has $\lambda_\ell - 1 = \lambda_1 - \ell = \lambda_s$ (since $\lambda_\ell = \lambda_1 - \ell + 1 = \lambda_s = \ell + 1$), so the parts are no longer distinct.

These two families correspond to the following shapes of Ferrers diagrams (with A in ●, B in ●, and the shared dot in ●):



One can verify the sizes directly. In case (1), the partition is $(2\ell-1,\ 2\ell-2,\ \dots,\ \ell)$, which has size

$$\sum_{i=\ell}^{2\ell-1} i = \frac{\ell(3\ell-1)}{2}.$$

In case (2), the partition is $(2\ell,\ 2\ell-1,\ \dots,\ \ell+1)$, which has size

$$\sum_{i=\ell+1}^{2\ell} i = \frac{\ell(3\ell+1)}{2}.$$

These are precisely the pentagonal numbers: writing $k = \ell$ gives $\frac{k(3k-1)}{2}$ for case (1), and writing $k = -\ell$ gives $\frac{(-\ell)(-3\ell-1)}{2} = \frac{\ell(3\ell+1)}{2}$ for case (2). In both cases the partition has ℓ parts, so its contribution to the signed count is $(-1)^\ell$.

Since the involution is parity-reversing on all other partitions with distinct parts (pairing them off in cancelling pairs), the net signed count of such partitions of n is exactly

$$\begin{cases} (-1)^\ell & \text{if } n = \frac{\ell(3\ell\pm1)}{2} \text{ for some } \ell \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

which completes the proof. □

§5.4 Jacobi triple product

Here are a couple of other identities that happen to be true:

Claim —

$$\prod_{n=1}^{\infty} (1 - x^n)^3 = \sum_k (-1)^k k x^{k(k+1)/2}.$$

Claim —

$$\prod_{n=1}^{\infty} \frac{1-x^n}{1+x^n} = \sum_k (-1)^k x^{k^2}.$$

It turns out these are all special cases of the following identity:

Theorem (Jacobi triple product)

$$\prod_{n=1}^{\infty} \left((1-x^{2n})(1+x^{2n-1}y^2)(1+x^{2n-1}/y^2)\right) = \sum_{n=-\infty}^{\infty} x^{n^2} y^{2n}.$$

For instance, we recover Euler's pentagonal number formula from $x = q^{3/2}, y^2 = -q^{1/2}$.

Proof. Reparameterize to z, q such that $x^2 = q$ and $y^2 = zq^{1/2}$; we wish to show that

$$\prod_{n=1}^{\infty} \left((1-q^n)(1+zq^n)(1+q^{n-1}/z)\right) = \sum_{n=-\infty}^{\infty} z^n q^{n(n+1)/2},$$

which we rearrange into

$$\left(\prod_{n=1}^{\infty}(1+zq^n)\right)\left(\prod_{n=1}^{\infty}(1+q^{n-1}/z)\right) = \left(\prod_{n=1}^{\infty}\frac{1}{1-q^n}\right)\left(\sum_{n=-\infty}^{\infty} z^n q^{n(n+1)/2}\right).$$

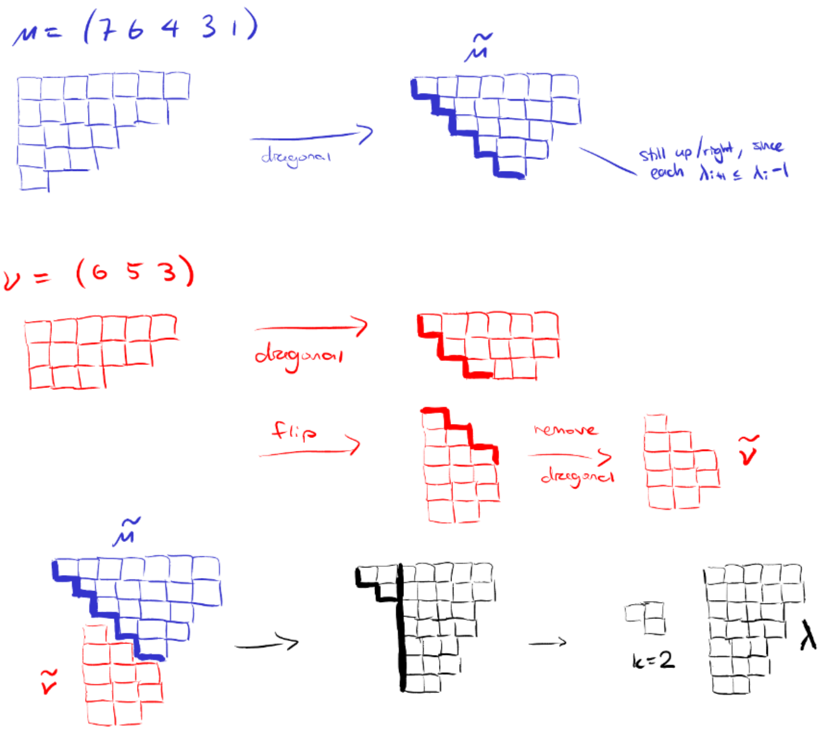
Let us first consider the LHS:

- the z^a coefficient of $\prod_{n=1}^{\infty}(1+zq^n)$ is the $\sum_\mu q^{|\mu|}$, where the sum is taken over all partitions μ with a parts, all distinct.
- analogously, the z^{-b} coefficient of $\prod_{n=1}^{\infty}(1+q^{n-1}/z)$ is $\sum_\nu q^{|\nu|-b}$, where the sum is taken over all partitions ν with b parts, all distinct.

Combining these, the $q^\ell z^k$ coefficient of the LHS is the number of pairs of partitions (μ, ν) such that μ has a parts all distinct, ν has b parts all distinct, $|\mu| + |\nu| - b = \ell$, and $a - b = k$.

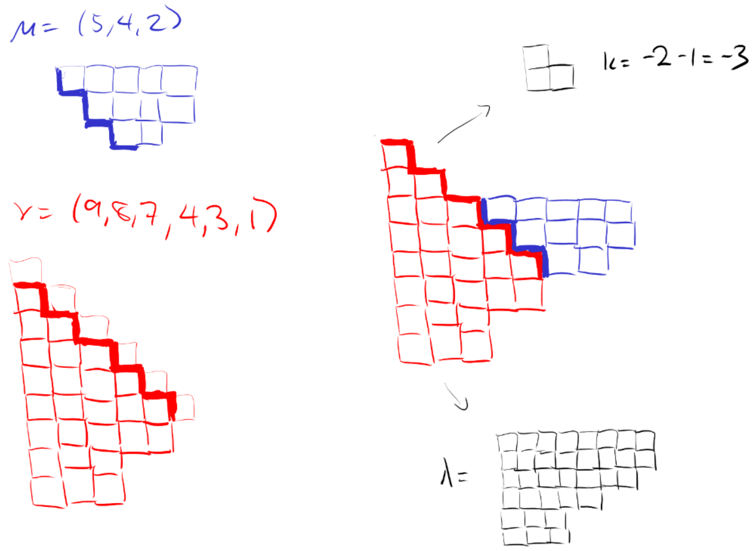
Meanwhile, on the RHS, $\prod_{n=1}^{\infty}\frac{1}{1-q^n}$ is just the generating function for partitions. So the $q^\ell z^k$ coefficient of the RHS is the number of partitions λ having size $|\lambda| = \ell - \frac{k(k+1)}{2}$. We will show that the LHS and RHS are equal by giving a bijection between such (μ, ν) and such (k, λ).

Case 1: $a \geq b$ (so $k = a - b \geq 0$):



We can check $a - b = k$ by simply looking at the height of the overlapping staircase region, and we can see that $|\mu| + |\nu| - b = \ell - \frac{k(k+1)}{2} = |\lambda|$ by counting the number of cells in λ.

Case 2: $a < b$ (so $k = a - b < 0$), we can do a similar process:



Correctness can be checked much as before. $\square$

Cutting and pasting like the argument above in different ways lets you prove similar claims, such as

$$\prod_{n \geq 1} (1 + xq^n) = \sum_{k=-\infty}^{\infty} \frac{q^{k(k+1)/2} x^k}{(1-q)(1-q^2)\cdots(1-q^k)}.$$

§5.5 Young's lattice

Consider the poset of partitions where the $\le$ operator is the subset. This poset is known as **Young's lattice**. The rank of a partition λ is imply $|\lambda|$; the rank numbers $r_n = p(n)$.

Note that f_λ, the number of SYT's of shape λ, is simply the number of saturated chains in Young's lattice from $\emptyset$ to λ; each chain gives us a labeling of the boxes of λ in some order into a SYT. Thus, $\sum_{|\lambda|=n} f_\lambda^2 = n!$ is the number of paths in the Hasse diagram that begin at $\emptyset$, move up n steps, and then move down n steps (thus ending back at $\emptyset$).

Let us write this idea out algebraically. Imagine a vector space V over $\mathbb{R}$, where for every YT λ, the vector space has a basis vector v_λ. Then for each integer $n \ge 0$ let V_n be the subspace of V spanned by all v_λ where $|\lambda| = n$. Of course, we have

$$V = V_0 \oplus V_1 \oplus V_2 \oplus \cdots .$$

Now we define an "up" operator U and a "down" operator D on V. To do this, it suffices to say how U and V act on each basis vector v_λ:

- $U(v_\lambda) = \sum_{\mu \leftarrow \lambda} v_\mu$, where the sum is taken over all μ obtained by adding one box to λ;
- $D(v_\lambda) = \sum_{\mu \rightarrow \lambda} v_\mu$, where the sum is taken over all μ obtained by removing one box from λ. Also we define $D(v_\emptyset) = 0$.

From the above, we now see that showing $\sum_{|\lambda|=n} f_\lambda^2 = n!$ is equivalent to showing $D^n U^n(v_\emptyset) = n! v_\emptyset$. Let us prove this result.

Lemma

In the Young's lattice,

$$DU - UD = I.$$

Proof. This is equivalent to showing that Young's lattice has the following two properties:

1. For any λ, the number of $\mu \leftarrow \lambda$ is one more than the number of $\mu \rightarrow \lambda$.
2. For any $\lambda \ne \mu$ with the same size, the number of $\nu \rightarrow \lambda, \mu$ and $\nu \leftarrow \lambda, \mu$ are the same.

□

Theorem

$$D^n U^n(v_\emptyset) = n! v_\emptyset.$$

Proof by induction. We first claim that for all n,

$$DU^n(v_\emptyset) = nU^{n-1}(v_\emptyset).$$

Induct on n. The base cases $n = 0, 1$ are easy. Consider $n \ge 2$:

$$DU^n(v_\emptyset) = (DU)U^{n-1}v_\emptyset = (UD + I)U^{n-1}(v_\emptyset) = UDU^{n-1}v_\emptyset + U^{n-1}v_\emptyset.$$

By the inductive hypothesis $DU^{n-1}v_\emptyset = (n-1)U^{n-2}v_\emptyset$, and we are done.

Now we can prove the original theorem by induction on n. Again $n = 0, 1$ are easy. Now for $n \geq 2$

$$D^n U^n(v_\emptyset) = D^{n-1} D U^n(v_\emptyset) = D^{n-1} n U^{n-1}(v_\emptyset) = n D^{n-1} U^{n-1}(v_\emptyset).$$

By the inductive hypothesis $D^{n-1}U^{n-1}(v_\emptyset) = (n-1)! v_\emptyset$ and we are done. $\square$

Another proof. A somewhat more enlightening proof is to consider the (linear) operators over the (linear) space of smooth functions

$$f(x) \to \frac{d}{dx} f(x)$$

and

$$f(x) \to x f(x).$$

Note that

$$\frac{d}{dx}(x f(x)) - x \frac{d}{dx} f(x) = f(x),$$

so these operators satisfy

$$\frac{d}{dx} x - x \frac{d}{dx} = \mathrm{id}.$$

And we of course know that $\frac{d^n}{dx^n} x^n = n!$. $\square$

A **differentiable poset** is a poset with unique minimum $\hat{o}$ where the up operator U and the down operator D satisfy $DU - UD = I$, i.e. it satisfies the two properties

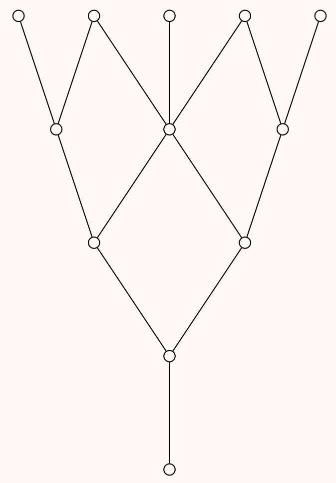
1. For any λ, the number of $\lambda \lessdot \mu$ is one more than the number of $\mu \to \lambda$.
2. For any $\lambda \neq \mu$ with the same size, the number of $\nu \lessdot \lambda, \mu$ and $\lambda, \mu \lessdot \nu$ are the same.

Our proof above really generalizes to show that for any differentiable poset,

$$\sum_{x \text{ with rank } n} \#(\text{saturated chain from } \hat{o} \text{ to } x)^2 = n!.$$

Example (Fibonacci lattice)

We can define the (differentiable) **differential poset** $\mathbb{F}$:



The ranks 0 and 1 both only have one element. For $k \geq 2$, the elements of rank k is obtained from the elements of rank $k-1$ and $k-2$:

- For each x of rank $k-1$, there exists an x' of rank k such that $x \lessdot x'$ (and this is the only relation $\bullet \lessdot x$).
- For each y of rank $k-2$ there exists y' of rank k such that for all x of rank $k-1$, we have $x \lessdot y'$ iff $y \lessdot x$.

§5.6 Oscillating tableaus

An **oscillating tableau** is a sequence of YTs $\lambda_0, \ldots, \lambda_{2n}$, starting and ending at $\emptyset$, where each YT is formed from the last by either adding or deleting a box. Note that we can associate an oscillating tableau with its sequence of $2n$ up-steps and down-steps. Note that, as this sequence can never go below $\emptyset$, it forms a Dyck path. This in turn can be viewed as a partition $\lambda \subseteq n \times n$ for which $\lambda_{n-i} > i$ for all $i = 0, \ldots, n-1$.

Claim — The number of oscillating tableaus with up/down steps corresponding to a given Dyck path D is the same as the number of ways to place non-attacking rooks in the region of the $n \times n$ square defined by D.

Proof sketch. The count of both, f_w, satisfies the following recurrence relation: If $w = w_1 U D w_2$, then let $w' = w_1 D U w_2$ and $w'' = w_1 w_2$;

$$f_w = f_{w'} + f_{w''}.$$

(This recurrence relation actually terminates because the YTs w', w'' are both smaller than w.) $\square$

With this equivalence, we can easily count the number to be $\prod_{i=0}^{n-1} (\lambda_{n-i} - i)$. Note that there must be exactly one rook in each row, and all rooks must be in different columns. Then

- there are λ_n ways to place the rook on the nth row of the partition;
- regardless of the the position of the nth rook, there are $\lambda_{n-1} - 1$ ways to place the rook on the $(n-1)$th row of the partition;
- regardless of the position of the nth and $(n-1)$th rooks, there are $\lambda_{n-2} - 2$ ways to place the rook on the $(n-2)$th row of the partition;
- and so on.

§6 Spanning trees

Theorem (Cayley)
The number of labelled trees on n vertices is n^{n-2}.

§6.1 Proof via algebra

Theorem (generalized Cayley)
Let

$$F(x_1, \dots, x_n) = \sum_T \prod_{i=1}^{n} x_i^{\deg_T(i)-1},$$

where the sum is taken over n-vertex labelled trees T.

Proof. Note that F_n is a homogenous (multivariable) polynomial of degree $n-2$, because all trees T have $\sum_{i=1}^n \deg_T(i) = 2(n-1)$, so $\sum_{i=1}^n (\deg_T(i) - 1) = n-2$. Let

$$G_n(x_1, \dots, x_n) = F_n - (x_1 + \dots + x_n)^{n-2},$$

so we wish to prove that $G_n \equiv 0$. We will do this by induction; this is obvious for $n \le 2$.

The following lemma will be useful:

Lemma
If $g(x_1, \dots, x_n)$ is a polynomial of degree $< n$ such that $g(x_1, \dots, x_n)|_{x_i=0} \equiv 0$ for all i, then $g \equiv 0$.

Proof. This is pretty simple: suppose otherwise, so that g contains some monomial $\prod_{i=1}^n x_i^{a_i}$. We have $a_1 + \dots + a_n < n$, so at least one $a_j = 0$; then $g(x_1, \dots, x_n)|_{x_j=0} \equiv 0$. □

Let us proceed with the proof. $G_n(x_1, \dots, x_n)$ is symmetric, so by the lemma, it would suffice to show that

$$G_n(x_1, \dots, x_{n-1}, 0) = \sum_T x_1^{\deg_T(x)-1} \cdots x_{n-1}^{\deg_T(n-1)-1} 0^{\deg_T(n)-1} \equiv 0.$$

The only nonzero terms in this sum are over the trees where n is a leaf, so that $\deg_T(n) - 1 = 0$. We can decompose these trees into n and a tree T' on vertices $1, \dots, n-1$, where n is connected to vertex $1 \le j \le n-1$. Now we have

$$\begin{aligned} G(x_1, \dots, x_{n-1}, 0) &= \sum_{T', j} x_j \prod_{k=1} x_k^{\deg_{T'}(k)-1} - (x_1 + \dots + x_{n-1})^{n-2} \\ &= \left(\sum_j x_j\right) \left(\sum_{T'} \prod_{k=1}^{n-1} x_k^{\deg_{T'}(k)-1}\right) - (x_1 + \dots + x_{n-1})^{n-2}. \end{aligned}$$

By the inductive hypothesis,

$$\sum_{T'} \prod_{k=1}^{n-1} x_k^{\deg_{T'}(k)-1} = (x_1 + \dots + x_{n-1})^{n-3},$$

and it follows that
$$G(x_1, \dots, x_{n-1}, 0) \equiv 0,$$
so $G \equiv 0$. □

This generalized version of Cayley's Theorem has some useful corollaries. For example, for some set of degrees $d_1, \dots, d_n \geq 1$, such that $d_1 + \cdots + d_n = n - 1$, the number of n-vertex labelled trees T where $\deg_T(i) = 1$ is the $x_1^{d_1-1} x_2^{d_2-1} \cdots x_n^{d_n-1}$ coefficient of $(x_1 + \cdots + x_n)^{n-2}$, which is $\binom{n-2}{d_1-1, d_2-1, \dots, d_n-1}$.

§6.2 Proof via Prufer codes

Seeing Cayley's formula above makes us wonder if there is a natural bijection between labelled trees and the set of sequences $[n]^{n-2}$. Indeed, this is given by **Prufer codes**:

> **Algorithm** — Given a labelled tree T with labels in $[n]$, produce a sequence in $[n]^{n-2}$ as follows: choose the leaf ℓ with the smallest label; let its parent be c. Then, append c to the sequence, remove ℓ from the tree, and repeat. When there are only two vertices left (connected by a single edge), terminate and return the resulting sequence.

Note x appears $\deg_T(x) - 1$ times in the Prufer code of tree T. More generally this property holds for all suffixes of the sequence, i.e. if we let T' be the tree with k vertices remaining, then code(T') is the $(k-2)$-length suffix of code(T), and the number of times x appears in code(T') is $\deg_{T'}(x) - 1$. This fact allows us to give the inverse mapping from Prufer codes back to trees:

> **Algorithm** — Given a Prufer code $c_1, c_2, \dots, c_{n-2}$, initialize C to be the sequence $[c_1, \dots, c_{n-2}]$ ("code"), and L to be the set of all labels not in the code ("usable labels") Now repeat $n-2$ times: take the first element c of C, take the minimal $\ell \in L$, and add the edge (ℓ, c) to the tree. Then remove ℓ from L and truncate the first element of C. If there are no more occurrences of c in C, add c to L. In the end C will be empty and L will have two remaining elements; add those two as the final edge.

Note that the Prufer codes give us another proof that the number of n-vertex labelled trees where vertex i has degree d_i is $\binom{n-2}{d_1-1, \dots, d_n-1}$, since these trees correspond exactly to codes where the label i appears $d_i - 1$ times.

§6.3 Proof via another bijection

We biject labelled trees to functions $f : [n] \to [n]$ with $f(1) = 1, f(n) = n$.

Given tree T, take the path from 1 to n in T; suppose this path is $a_1, a_2, \dots, a_k$ with $a_1 = 1$ and $a_k = n$. Find the a_i which are the smallest records when the sequence is read left-to-right, i.e. $a_i < a_j$ for all $j > i$. (Note that a_k is a record, as is a_1.) Then we can partition the path into blocks $\{a_{i_1 = 1}\}, \{a_{i_1+1}, \dots, a_{i_2}\}, \{a_{i_2+1} \cdots a_{i_3}\}, \dots$, where the rightmost vertex of each block is a record.

Now define f:

- if x is not on the path, define $f(x)$ to be the neighbor of x in the direction towards the path.
- if $x = a_j$ is on the path, and *not* the leftmost element of a block, then define $f(a_j) = a_{j-1}$.
- if $x = a_j$ is on the path, and is the leftmost element of the block associated with record a_{i_p}, define $f(a_j) = a_{i_p}$.

§6.4 Proof via Kirchoff's Matrix Tree Theorem

A **spanning tree** of a (multi)graph G is a subset of the edges of G that form a tree on all the vertices of G. In this section we aim to count the number of spanning trees of a graph.

For a graph G with n vertices and m edges, define the following matrices:

- The **adjacency** matrix $A \in \mathbb{R}^{n\times n}$, where A_{ij} is the number of edges between i and j.
- The **incidence** matrix $C \in \mathbb{R}^{n\times m}$, where C_{ie} is 1 if i is an endpoint of edge e, and 0 otherwise. note that $A = CC^\intercal$.
- The **degree** matrix $D \in \mathbb{R}^{n\times n}$, the diagonal matrix with $D_{i,i}$ equal to the degree of vertex i;
- The **Laplacian** matrix $L \in \mathbb{R}^{n\times n}$, given by $L = D - A$; note that $\det L = 0$ because all rows (and columns) of L sum to zero.
- The **reduced Laplacian** matrix $\tilde{L} \in \mathbb{R}^{(n-1)\times(n-1)}$, given by removing the last row and column from L.

Theorem (Kirchoff's Matrix Tree Theorem)
The number of spanning trees of a graph is equal to the determinant of its reduced Laplacian.

Proof. Define the **oriented incidence matrix** B: orient every edge in the graph arbitrarily, and then define $B \in \mathbb{R}^{n\times m}$ such that

$$B_{i,e} = \begin{cases} 1 & \text{if } i \text{ is the source of edge } e \\ -1 & \text{if } i \text{ is the sink of } e \\ 0 & \text{otherwise} \end{cases}.$$

The dot product of any row B_i with itself is

$$B_i \cdot B_i = \sum_e (\pm 1[i \text{ is endpoint of } e])^2 = \deg_i.$$

The dot product of any two rows $B_i \neq B_j$ (for $i \neq j$) is

$$B_i \cdot B_j = \sum_e -1[i, j \text{ are the endpoints of } e].$$

so $BB^\intercal = L$. Letting $\tilde{B} \in \mathbb{R}^{(n-1)\times m}$ be the "reduced" oriented incidence matrix with its last row truncated, we can see that $\tilde{L} = \tilde{B}\tilde{B}^\intercal$.

Lemma (Cauchy-Binet)

If $C, D \in \mathbb{R}^{k\times m}$ then

$$\det(CD^\intercal) = \sum_{\substack{S\subset[m]\\|S|=k}} \det(C_S)\det(D_S),$$

where C_S and D_S refer to the $k\times k$ matrices formed by sampling columns of C and D according to the indices of S.

(We will not prove this lemma; you can find a proof on Wikipedia.)

Citing Cauchy-Binet on our expression we wish to find

$$\det(\tilde{L}) = \det(\tilde{B}\tilde{B}^\intercal) = \sum_{\substack{S\subset[m]\\|S|=n-1}} \det(\tilde{B}_S)\det(\tilde{B}_S^\intercal) = \sum_{\substack{S\subset[m]\\|S|=n-1}} \det(\tilde{B}_S)^2.$$

We wish to show the right-hand sum is just the number of spanning trees. This will be done by showing the summation terms are exactly 1 when S is spanning trees and 0 otherwise:

Lemma

For $S \subset [m]$ with $|S| = n-1$, the determinant $\det(\tilde{B}_S)$ is ± 1 if S corresponds to the edges of a spanning tree, and $\det(\tilde{B}_S) = 0$ otherwise.

> **Remark.** In fact, one can show that any oriented incidence matrix B of a graph is **totally unimodal**, i.e. all its minors are one of $-1, 0, 1$.

Proof. Suppose S does not form a spanning tree; we will show $\det(\tilde{B}_S) = 0$. Since S is a set of $n-1$ edges, but is not a tree, the subgraph spanned by S must contain a cycle. Let the edges of the cycle be $e_1, e_2, \dots, e_k$ (of course, $k \le n-1$); note that the sum of the columns

$$B_{e_1} \pm B_{e_2} \pm \cdots \pm B_{e_k} = 0,$$

where the signs on the $\pm$'s are chosen to orient all edges of the cycle consistently (with the direction of edge e_1).

Then, as $\tilde{B}$ truncates a row from B, it is of course also true that the sum of these columns in $\tilde{B}$ is $\sum_{i=1}^{k} \pm\tilde{B}_{e_i} = 0$. This implies the columns $\{e_1, \dots, e_k\} \subseteq S$ are not linearly independent, so $\tilde{B}_S$ does not have full rank and its determinant is indeed zero.

Now suppose S does form a spanning tree. Consider that in B_S, row i only has one nonzero entry (which must be ± 1) if and only if it is a leaf. Now, we induct on n: the base case $n = 1$ is trivial. For the inductive step, assume $n \ge 2$: our tree S has at least two leaves, which means one of them is $i \ne n$ (recall that the last vertex is being truncated from B to $\tilde{B}$). Thus the row $\tilde{B}_i$ contains only a single ± 1 entry in column e.

Let S' be the tree with vertices $[n] \setminus i$ and edges $S \setminus e$; its oriented adjacency matrix B' is just B with the ith row and eth column removed. Of course then $\tilde{B}'$ is $\tilde{B}$ with ith row and eth column removed. By the inductive hypothesis we have $\det(\tilde{B}') = \pm 1$, and by row reduction on B we have

$$\det(\tilde{B}) = B_{ie}\det(\tilde{B}') = \pm 1.$$

$\square$

This completes the proof of Kirchoff's Matrix Tree Theorem. $\square$

> **Remark.** This generalizes to weighted graphs by taking the weighted Laplacian, and summing over the weights of each spanning tree (the weight of a tree is the product of the weights of its edges).

The matrix tree theorem can also be written in a purely spectral form:

Theorem

If the eigenvalues of L are $\lambda_1, \ldots, \lambda_{n-1}, \lambda_n$ with $\lambda_n = 0$, then the number of spanning trees of the graph is

$$\frac{1}{n}\prod_{i=1}^{n-1}\lambda_i.$$

Proof. Leibniz formula on $\det(\lambda I - L)$ gives that the λ^{n-k} term is $(-1)^k$ times the sum of the $k \times k$ principal minors. When $k = n-1$, we get that the λ^1 term is $(-1)^{n-1}$ times the sum of all the determinants of the first minors $\det \tilde{L}_i$. There are n of these minors, each equal to the number of spanning trees $\tau(G)$. Meanwhile, $\det(\lambda I - L) = \prod(\lambda - \lambda_i)$. Recalling $\lambda_n = 0$, the λ term is $(-1)^{n-1}\prod_{i=1}^{n-1}\lambda_i$. Putting these two together, $n\tau(G) = \prod_{i=1}^{n-1}\lambda_i$ as desired. $\square$

We can now deduce Cayley's Formula from Kirchoff's Matrix Tree Theorem applied to the complete graph K_n; the Laplacian is

$$L = nI_n - \mathbf{1}_{n\times n}$$

so the reduced Laplacian is

$$\tilde{L} = nI_{n-1} - \mathbf{1}_{(n-1)\times(n-1)}.$$

Note that the eigenvalues of $\mathbf{1}_{(n-1)\times(n-1)}$ are $n-1, 0, \ldots, 0$, so the eigenvalues of $\tilde{L}$ are $1, n, \ldots, n$, and thus $\det(\tilde{L})$ is the product of the eigenvalues, which is n^{n-2}.

§6.5 Counting spanning trees of the hypercube

Call the **spectrum** of a graph the eigenvalues $\alpha_1, \ldots, \alpha_n$ of its adjacency matrix. These in general have nothing to do with the eigenvalues $\lambda_1, \ldots, \lambda_n$ of the Laplacian matrix (and thus the matrix tree theorem). However, when the graph is **d-regular**, i.e. all vertices have degree d, we have

$$L = dI - A$$

so the spectrum $\alpha_i = d - \lambda_i$. The number of spanning trees of a d-regular graph can therefore be computed from the spectrum as $\frac{1}{n}\prod_{i=1}^{n-1}(d - \alpha_i)$.

Definition. For graphs G and H, define their **Cartesian product** $G \times H$ as the graph on vertex set $V(G) \times V(H)$ with the following edges: for $(u, v), (u', v') \in V(G) \times V(H)$, we have $(u, v) - (u', v')$ iff either

- $u = u'$ and $vv' \in E(H)$, or
- $uu' \in E(G)$ and $v = v'$.

The spectrum has the following property with respect to the Cartesian product:

Lemma

If G has spectrum $\alpha_1, \ldots, \alpha_m$, and H has spectrum $\beta_1, \ldots, \beta_n$, then $G \times H$ has spectrum $\{\alpha_i + \beta_j\}_{i,j \in [m]\times[n]}$.

> **Remark.** We can have eigenvalues with multiplicity here, so all sets of eigenvalues here should be interpreted as multisets.

Proof. Let the eigenvectors of the adjacency matrices of G, H be $u_1, \ldots, u_m$ and $v_1, \ldots, v_n$. Note that the adjacency matrix is symmetric, so by the Spectral Theorem we can guarantee orthonormal eigenbases and such eigenvectors can be found.

For vectors $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$, define their **Kronecker (tensor) product** $x \otimes y \in \mathbb{R}^{mn}$ is the vector indexed by pairs $(a, b) \in [m] \times [n]$ with entries

$$(x \otimes y)_{(a,b)} = x_a \cdot y_b.$$

For matrices $A \in \mathbb{R}^{m_1 \times m_2}$ and $B \in \mathbb{R}^{n_1 \times n_2}$, the tensor product $A \otimes B \in \mathbb{R}^{m_1 n_1 \times m_2 n_2}$ is the matrix indexed by pairs with entries

$$(A \otimes B)_{(a,b),(c,d)} = A_{a,c} \cdot B_{b,d}.$$

We first claim that matrices $A \in \mathbb{R}^{m_1 \times m_2}$, $B \in \mathbb{R}^{n_1 \times n_2}$, $C \in \mathbb{R}^{m_2 \times m_3}$, $D \in \mathbb{R}^{n_2 \times n_3}$,

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

Indeed, the $((a, b), (e, f))$-entry of the left-hand side is

$$\begin{aligned}
\sum_{(c,d)} (A \otimes B)_{(a,b),(c,d)}(C \otimes D)_{(c,d),(e,f)} &= \sum_{c,d} A_{a,c}B_{b,d}C_{c,e}D_{d,f} \\
&= \left(\sum_c A_{a,c}C_{c,e}\right)\left(\sum_d B_{b,d}D_{d,f}\right) \\
&= (AC)_{a,e}(BD)_{b,f},
\end{aligned}$$

which is the $((a,b),(e,f))$-entry of $(AC)\otimes(BD)$.

Now, recall that the vertex set of $G\times H$ is $V(G)\times V(H)$, where $(g,h)\sim(g',h')$ if and only if either $g=g'$ and $h\sim h'$, or $g\sim g'$ and $h=h'$. Reading off the adjacency matrix entry at $((g,h),(g',h'))$:

$$(A_{G\times H})_{(g,h),(g',h')} = (A_G)_{g,g'}\delta_{h,h'} + \delta_{g,g'}(A_H)_{h,h'} = (A_G\otimes I_n + I_m\otimes A_H)_{(g,h),(g',h')},$$

so $A_{G\times H} = A_G\otimes I_n + I_m\otimes A_H$.

We claim that $u_i\otimes v_j$ is an eigenvector of $A_{G\times H}$ with eigenvalue $\alpha_i+\beta_j$. By the mixed-product property:

$$\begin{aligned}
A_{G\times H}(u_i\otimes v_j) &= (A_G\otimes I_n)(u_i\otimes v_j) + (I_m\otimes A_H)(u_i\otimes v_j)\\
&= (A_G u_i)\otimes(I_n v_j) + (I_m u_i)\otimes(A_H v_j)\\
&= \alpha_i(u_i\otimes v_j) + \beta_j(u_i\otimes v_j)\\
&= (\alpha_i+\beta_j)(u_i\otimes v_j).
\end{aligned}$$

It remains to show that these mn eigenvectors form a basis (to check that we have not accidentally constructed some duplicate eigenvectors). Since $\{u_i\}$ and $\{v_j\}$ are orthonormal, we have

$$\begin{aligned}
\langle u_i\otimes v_j,\, u_k\otimes v_\ell\rangle &= \sum_{(a,b)}(u_i)_a(v_j)_b(u_k)_a(v_\ell)_b\\
&= \left(\sum_a (u_i)_a(u_k)_a\right)\left(\sum_b (v_j)_b(v_\ell)_b\right)\\
&= \langle u_i,u_k\rangle\langle v_j,v_\ell\rangle\\
&= \delta_{ik}\,\delta_{j\ell}.
\end{aligned}$$

Thus $\{u_i\otimes v_j\}_{(i,j)\in[m]\times[n]}$ is an orthonormal set of mn vectors in $\mathbb{R}^{mn}$, hence a basis. We have found all mn eigenvalues, so the spectrum of $G\times H$ is $\{\alpha_i+\beta_j\}_{(i,j)\in[m]\times[n]}$. □

We can use this to find the number of spanning trees in a hypercube, by noting that if $H_n = H_1\times\cdots\times H_1$, where H_1 is a single edge, and thus has adjacency matrix

$$A(H_1) = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix},$$

which has eigenvalues $\{-1,1\}$. Thus the adjacency matrix $A(H_n)$ has eigenvalue set $\{-1,1\}+\cdots+\{-1,1\}$, i.e. the eigenvalue $d-2k$ appears with multiplicity $\binom{d}{k}$ for $0\le k\le d$. The Laplacian thus has eigenvalue $2k$ appearing with multiplicity $\binom{d}{k}$ for $0\le k\le d$, so the number of spanning trees is then

$$\frac{1}{v(H_n)}\prod_{0<k\le d}(2k)^{\binom{d}{k}} = 2^{2^d-d-1}\prod_{k=1}^{d}k^{\binom{d}{k}}.$$

§6.6 A polynomial related to spanning trees

For graph G with vertices $1, \dots, n$, define G^+ to be the graph obtained by adding a vertex 0 to G that is connected to all other vertices. Then define the polynomial

$$f_G(x) = \sum_T x^{\deg_T(0)-1},$$

where the sum is taken over all spanning trees T of G^+.

- since $\deg_T(0)$ is at most n, and there is only one spanning tree (the star at 0) of G^+ where $\deg_T(0) = n$, the x^{n-1} coefficient is 1.
- meanwhile, the constant term (terms where $\deg_T(0) = 1$) is n times the number of spanning trees of G: we get one term per spanning tree of G and choice of parent of 0.

Theorem (reciprocity)
If $\bar{G}$ is the complement of G,

$$f_{\bar{G}}(x) = (-1)^{n-1} f_G(-x-n).$$

Proof. Generalize $f_G(x)$ to

$$F_G(x, x_1, \dots, x_n) = \sum_{\text{spanning tree } T \text{ of } G^+} x^{\deg_T(0)-1} \prod_{i=1}^{n} x_i^{\deg_T(i)-1}.$$

We will show that

$$F_{\bar{G}}(x, x_1, \dots, x_n) = F_G(-x-n, x_1, \dots, x_n).$$

Imagine x is constant and consider F_G and $F_{\bar{G}}$ to be polynomials in $x_1, \dots, x_n$; we can use an inductive argument very similar to the one in Cayley's formula where we show that $F_{\bar{G}}(x, x_1, \dots, x_n)|_{x_i=0} = F_G(-x-n, x_1, \dots, x_n)|_{x_i=0}$ for all $1 \le i \le n$, then conclude $F_G(x, x_1, \dots, x_n) \equiv F_{\bar{G}}(x, x_1, \dots, x_n)$. $\square$

Lemma

$$f_{G \sqcup H}(x) = x f_G(x) f_H(x).$$

Proof. This follows immediately from noting that each spanning tree of $(G \sqcup H)^+$ decomposes into a spanning tree of G^+ and a spanning tree of H^+. $\square$

Applying this lemma allows us to compute the polynomial f_G for various classes of graphs which are highly symmetric, and thus count the number of spanning trees. For example, we can begin with the polynomial of the one-vertex graph

$$f_\bullet(x) = 1,$$

and find that the polynomial of the independent graph

$$f_{\bar{K_n}}(x) = x^{n-1},$$

so the polynomial of the complete graph is

$$f_{K_n}(x) = (-1)^{n-1} f_{\bar{K_n}}(-x-n) = (-1)^{n-1}(-x-n)^{n-1} = (x+n)^{n-1}.$$

From here we can recover Cayley's formula by looking at the constant term of the polynomial. Note that it is actually possible to further continue this argument: the polynomial

$$f_{K_n \sqcup K_m}(x) = x(x+m)^{m-1}(x+n)^{n-1},$$

and by taking the complement, the polynomial of the complete bipartite graph is

$$f_{K_{m,n}}(x) = (x+m+n)(x+n)^{m-1}(x+m)^{n-1}.$$

and thus by looking at the constant term, $K_{m,n}$ has $n^{m-1}m^{n-1}$ spanning trees.

§6.7 Directed matrix tree theorem

Here we find a generalization of the matrix tree theorem to directed graphs. Consider digraph $G = (V, E)$; we can write its adjacency matrix A where

$$a_{ij} = \#\text{edges from } i \text{ to } j,$$

and let its Laplacian be $L = D - A$, where D is the diagonal matrix of *indegrees*.

Define an **arborescence** at vertex $r \in V$ to be a spanning tree, rooted at r, where all $n-1$ edges of the tree point from parent to child (outwards from r).

Theorem

Recall that the cofactor $L^{k,r}$ is given by $(-1)^{k+r} \det C_{k,r}$, where $C_{k,r}$ is the minor obtained by removing the kth row and rth column from L. Then $L^{k,r}$ is constant over all values of k, and is the number of arborescences at r, denoted $\operatorname{Arb}_r(G)$.

Proof. First, assume G has no edges pointing towards r. If they exist, we can simply delete them. This does not change $\operatorname{Arb}_r G$, because no arborescence at r contains an edge pointing towards r; it also does not change $L^{k,r}$, because edges $i \to r$ correspond to coefficients in the rth column of L, which are deleted in the minor $C_{k,r}$.

Now, after making this assumption, let us induct on the number of edges of the graph. The base case, where G has no edges, is obvious; the number of arborescences is zero, and $L^{k,r} = 0$.

For our inductive step, consider arbitrary G; consider some edge $e = (i \to j)$ in the graph. Since there are no edges pointing towards r, we know $j \neq r$. Now, consider the two graphs:

- G_1 consists of all edges of G, except for e.
- G_2 consists of all edges of G, except for all other edges pointing towards j.

Of course, $G_1 \subset G$ and $G_2 \subseteq G$. Suppose there exists some edge e for which $G_2 \subset G$. It is quite clear that the Laplacians L_{G_1}, L_{G_2}, and L_G are identical except for the jth column, where the sum of the jth columns of L_{G_1} and L_{G_2} equals the jth column of L_G. The same, of course, is thus true about the cofactors $C_{G_1}^{k,r}$, $C_{G_2}^{k,r}$, and $C_G^{k,r}$. By linearity of the determinant we conclude $L^{k,r} = L_{G_1}^{k,r} + L_{G_2}^{k,r}$. The inductive hypothesis tells us that $L_{G_1}^{k,r}$ is the number of arborescences at r in graph G_1, and that $L_{G_2}^{k,r}$ is the number of arborescences at r in graph G_2. Of course, any arborescence in G is either an arborescence in G_1 or G_2 (but not both), because the edge pointing to j in the arborescence must either be e, or not e. So indeed

$$\operatorname{Arb}_r(G) = \operatorname{Arb}_r(G_1) + \operatorname{Arb}_r(G_2) = L_{G_1}^{k,r} + L_{G_2}^{k,r} = L^{k,r}.$$

In the event that there is no edge e for which $G_2 \subset G$, the above argument does not hold, because the inductive hypothesis obviously cannot be applied to $G_2 = G$. In this case, all vertices $j \neq r$ must have indegree exactly 1. Recall that r itself has indegree zero, so G has exactly $n-1$ edges. Therefore, we only have two cases:

- G contains a loop, $v_1 \dots v_\ell$. Note that this loop is its own connected component, and does not contain r. So there are no arborescences at r. Meanwhile, the columns $v_1, \dots, v_\ell$ of the cofactor $C_{k,r}$ are linearly dependent, so $L^{k,r} = (-1)^{k+r} \det C_{k,r} = 0$.

- G is an arborescence. Then we saw earlier (in a lemma used in the proof of the matrix tree theorem) that $L^{k,r} = \pm 1$. It is possible to check the signs very carefully to see the signage is in fact positive.

□

Remark. The directed matrix tree theorem generalizes in the obvious way to weighted graphs.

§6.8 Electrical networks

In this section we show an interesting relation between electrical networks and spanning trees. Imagine an electrical network as a graph G with n nodes; each edge e is a resistor with resistance R_e.

Suppose now that vertex 1 is a voltage source of voltage 1, and vertex n is grounded at voltage 0. Kirchoff's circuit laws are equivalent to the existence of a **potential function** V at each vertex, which satisfies the conditions

- $V_1 = 1$ and $V_n = 0$;
- the current through each edge ij (from $i \to j$) is $\frac{V_j - V_i}{R_{ij}}$;
- for each vertex $j \neq 1, n$, the sum of the currents into each vertex is zero:

$$\sum_i \frac{V_j - V_i}{R_{ij}} = 0.$$

Define the conductances $x_e = \frac{1}{R_e}$, and let K be the weighted Laplacian matrix of the graph G where the edge weights are the conductances. With $V = \langle V_1, \dots, V_n \rangle$, the condition above gives us $KV = \langle J, 0, \dots, 0, -I \rangle$ for some current J.

Remove the last row and column K to get $\tilde{K}$, and remove the last index of V to get $\tilde{V}$; we have $\tilde{K}\tilde{V} = \langle J, 0, \dots 0 \rangle$.

Lemma (Cramer's rule)

Suppose x is the solution to $Ax = b$. Let A_i be the matrix identical to A except for the ith column, which has been replaced by b. Then

$$x_i = \det(A_i)/\det(A).$$

Applying Cramer's rule to the first coefficient here we see that

$$1 = V_1 = \frac{\det K'}{\det \tilde{K}}$$

where $\tilde{K}'$ is the matrix $\tilde{K}$, except the first column has been replaced with $\langle J, 0, \dots, 0 \rangle$. By row reduction, we have $\det \tilde{K}' = J \det \tilde{\tilde{K}}$, where $\tilde{\tilde{K}}$ is the result of removing both indices 1 and n from K. We can rewrite as

$$J = \frac{\det \tilde{K}}{\det \tilde{\tilde{K}}},$$

which means the equivalent resistance of the circuit is $R_{\text{eff}} = \frac{1}{J} = \frac{\det \tilde{\tilde{K}}}{\det \tilde{K}}$.

Note that by Kirchoff's matrix tree theorem, $\det \tilde{K}$ is the sum of the weights of the spanning trees of the original network G. Meanwhile, $\det \tilde{\tilde{K}}$ is the sum of the weights of the spanning trees of the graph $\tilde{G}$ where the voltage source (vertex 1) and the ground (vertex n) of G are shorted together, i.e. merged at the same potential into one vertex.

Example

If G is a series of two resistors with resistances R_1 and R_2, then G has only one spanning tree, which is G itself. The weight of this tree is $\frac{1}{R_1 R_2}$.

Meanwhile, $\tilde{G}$ is two parallel edges with resistances R_1 and R_2, it has two spanning trees, namely the two edges, with combined weight $\frac{1}{R_1} + \frac{1}{R_2}$.

The equivalent resistance of the circuit is thus

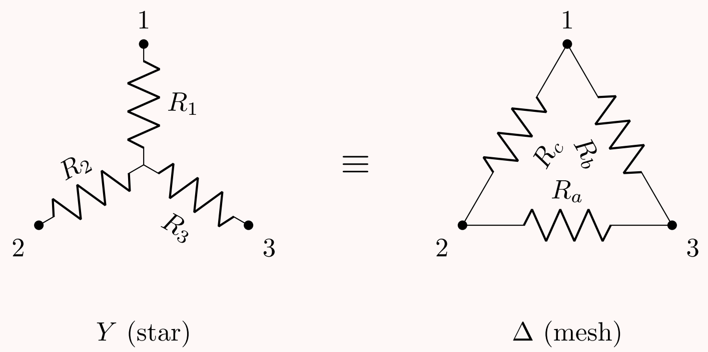
$$\frac{\frac{1}{R_1} + \frac{1}{R_2}}{\frac{1}{R_1 R_2}} = R_1 + R_2.$$

Example

If G is two resistors in parallel with resistances R_1 and R_2, it has two spanning trees, namely the two edges, with combined weight $\frac{1}{R_1} + \frac{1}{R_2}$. Meanwhile $\tilde{G}$ is a single node, so it has one spanning tree of weight 1. The equivalent resistance is thus $\frac{1}{1/R_1 + 1/R_2}$.

Example

Another useful trick in simplifying circuits that can be shown using Kirchoff's matrix tree theorem is the Y-delta transform, which states that the following two circuit elements are equivalent:



where $R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$ (and symmetrically for R_2 and R_3).

Another interesting interpretation of the electrical potential function arises from random walks on graphs. Consider a graph G with edge weights x_e, and consider a random walk on this graph where the probability of moving from vertex i to vertex j is $\frac{x_{ij}}{\sum_{j'} x_{ij'}}$. Then let $p(v)$ be the probability that, if this walk starts at vertex v, it reaches vertex 1 before vertex n. Then $p(1) = 1$, $p(n) = 0$, and for all other vertices v, we have

$$p(v) = \frac{\sum_j x_{vj} p(j)}{\sum_j x_{vj}},$$

which means p is exactly the electrical potential function V from before on a network with conductances x_e.

§6.9 Parking functions

Imagine n cars, labelled 1 through n, trying to park in n parking spots, also labelled 1 through n (from left to right). For some $f : [n] \to [n]$, each car in order attempts to park via the following strategy: the ith car proceeds directly to spot f_i, and then takes the first (leftmost) empty parking spot from there.

Call f a **parking function** if this results in all cars successfully parking.

Example

$(3, 3, 1, 1)$ is a parking function:

1. first car goes to spot 3 and parks;
2. second car goes to spot 3, finds it occupied, parks in spot 4;
3. third car goes to spot 1 and parks;
4. fourth car goes to spot 1, finds it occupied, parks in spot 2.

Meanwhile $(3, 3, 2, 2)$ is not a parking function:

1. first car goes to spot 3 and parks;
2. second car goes to spot 3, finds it occupied, parks in spot 4;
3. third car goes to spot 2 and parks;
4. fourth car goes to spot 2, finds all spots 2, 3, 4 occupied, is sad.

Lemma

$f : [n] \to [n]$ is a parking function iff for all k, there are at most k of the $f_i \geq n-k+1$.

Theorem

The number of parking functions is $(n + 1)^{n-1}$.

Proof. Modify the parking procedure to n cars and $n + 1$ spots in a circle; thus modified parking functions are $f : [n] \to [n+1]$. Let P_i be the set of all parking functions such that the ith spot is left empty at the end of the parking algorithm. Then our original parking functions correspond exactly to P_{n+1}; all P_i are symmetrical, and there are $(n+1)^n$ total parking functions, so $|P_{n+1}| = (n + 1)^{n-1}$. □

This expression should remind us of Cayley's formula. Before this, let us show give a bijection between parking functions and **labelled Dyck paths**:

Definition. A **labelled Dyck path** is an assignment of the numbers $1, \ldots, n$ to the up-steps of a Dyck path of length $2n$, such that labels increase in consecutive up-steps.

Consider the following algorithm to construct a labelled Dyck paths and parking functions: View the Dyck path as a sequence of up/right steps that never goes above the line $y = x$. Label the y-levels $1, \ldots, n$ from bottom to top. Now build the parking function: for each label i at y-level y, assign $f(i) = y$.

This is a bijection.

§7 BEST theorem

Suppose $G = (V, E)$ is an Eulerian graph: there exists a permutation of all the edges defining a walk that begins and ends at the same vertices.

Theorem

A (directed) graph G is Eulerian if and only if its undirected version is connected, and all indegrees are equal to all outdegrees.

Note that if $(e_1, \ldots, e_m)$ is an Eulerian cycle, so is $(e_2, \ldots, e_m, e_1)$; we can of course partition Eulerian cycles into equivalence classes up to rotation, each of size m.

Theorem

Suppose G is Eulerian. Then the number of Eulerian cycles of G is

$$|E| \cdot \mathrm{Arb}_r(G) \cdot \prod_{v \in V} (\mathrm{indeg}_G(v) - 1)!.$$

Note that this implies all $\mathrm{Arb}_r(G)$ are equal.